

Why Public-Safety Wages May Grow Differently

A corrected visual atlas of municipal compensation evidence

Compensation mechanisms, claim boundaries, project-wide limitations, and the complete Human–AI workflow

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Gabriel Wages · August 2026 · revised for handoff

What the evidence supports—and what it does not

The strongest result

The reviewed corpus supports a bounded account of compensation mechanisms and directional pressures. It supports one final claim, conditionally supports one, supports five as mechanisms only, treats one as mixed, and leaves six unsupported.

What changed in this revision

The map extent now includes the full lower 48 and Alaska; related mechanisms and claims are integrated; safety/non-safety visibility is explicit; and limitations and methodology cover the full project.

Contents

Part I explains terms. Part II describes the corpus. Part III integrates mechanisms and claims. Part IV compares patterns. Part V preserves all claim boundaries. Part VI pairs limits with accomplishments. Part VII documents the complete workflow.

What this project asked

Research question

Why might public-safety wages grow differently from other municipal wages within the same city and compensation cycle?

Design discipline

The intended analytical comparison holds municipality and time fixed while occupation varies. Safety units without a matched non-safety unit are not sufficient for a comparative wage estimate.

Final boundary

The corpus is strongest for how compensation is set, implemented, and constrained. It is not a representative national wage panel.

How to read the atlas

Start with the unit

Every count names its analytical unit. Sources, spans, events, claims, and municipalities are not interchangeable.

Read the boundary

Each mechanism profile states what the evidence supports and what it does not establish.

Treat imbalance descriptively

Safety records appear more often in classified evidence. This can reflect institutions, labels, source formats, and discovery—not only real-world frequency.

How to read a hex map

Why hexagons

A hex map groups nearby municipalities into equal-sized areas so regional patterns are easier to see.

What shading means

Darker areas contain more distinct documented implementation events. The fixed extent is identical across profiles.

What shading does not mean

It does not represent more workers, a larger wage effect, population prevalence, or national frequency. Alaska appears in a labeled inset.

Evidence tiers

Strict evidence

Exact, claim-safe source evidence with compatible units and reproducible formulas.

Bounded evidence

Compatible evidence with an explicit caveat; useful for bounded comparisons and sensitivity.

Directional evidence

Mechanism or direction without a precise magnitude.

Context and rejected evidence

Context explains the setting; rejected evidence cannot support a claim.

Safety and non-safety

Safety

Police, fire, and explicitly combined public-safety units.

Non-safety

Other municipal occupations. Their compensation is often embedded in broad budgets, personnel schedules, or multi-department records.

Other categories

Mixed records include both sides; side-independent records apply across or outside the distinction; unresolved records are not forced into a side.

What an implementation event is

Declared map unit

One root compensation event in one municipality and compensation cycle, attached to one compensation category and employee side.

Deduplication correction

The source layer contains 13,391 mechanism-link rows. Removing repeated family labels at the declared map unit yields 11,698 map events. The two counts describe different layers and are not added.

Scope

Only formal adoption, implementation, or observed payment records in the retained layer enter these maps.

Compensation and evidence glossary

Safety

Police, fire, or explicitly combined public-safety employees.

Non-safety

Other municipal occupations, including clerical, public works, library, parks, sanitation, health, and related roles.

Compensation mechanism

A process or rule that sets, changes, implements, or constrains pay.

Implementation event

A distinct municipality × compensation cycle × mechanism × side record showing formal adoption, implementation, or observed payment.

Compensation cycle

The contract, fiscal, calendar, or administrative period in which compensation terms apply.

Strict evidence

Exact, source-supported evidence with compatible units and reproducible calculations.

Bounded evidence

Analytically usable evidence with a stated role, period, source, or comparison caveat.

Directional evidence

Evidence about a mechanism or direction without a defensible precise magnitude.

Administrative record

A payroll, staffing, budget, ordinance, pay-plan, or implementation document.

Documentary evidence

Contracts, awards, agreements, ordinances, and reports.

Side-independent

A record that applies across employee sides or does not require a side.

Mixed

A record that explicitly involves both safety and non-safety employees.

Unresolved

A record whose side or mechanism cannot be assigned without unsupported inference.

Local comparison

A bounded same-municipality comparison that remains local, not national.

Compensation and evidence glossary — continued

Retroactive pay

Back pay owed because terms apply to an earlier period.

Pay plan

An administrative schedule or process setting compensation.

Step progression

Scheduled movement through pay steps.

Salary schedule

A table of rates, grades, ranges, or steps.

COLA

A cost-of-living adjustment, sometimes tied to inflation.

Classification adjustment

A change to job grade, band, or pay placement.

Across-the-board raise

The same general increase across covered classifications.

Compression

Narrow pay differences between ranks, steps, or related jobs.

Non-base pay

Compensation beyond the recurring base rate.

Premium pay

Additional pay for duties, qualifications, shifts, or conditions.

Bargaining leverage

Pressure employees or unions can use during negotiations.

Staffing pressure

Vacancy, recruitment, retention, or minimum-staffing conditions entering pay decisions.

PART II

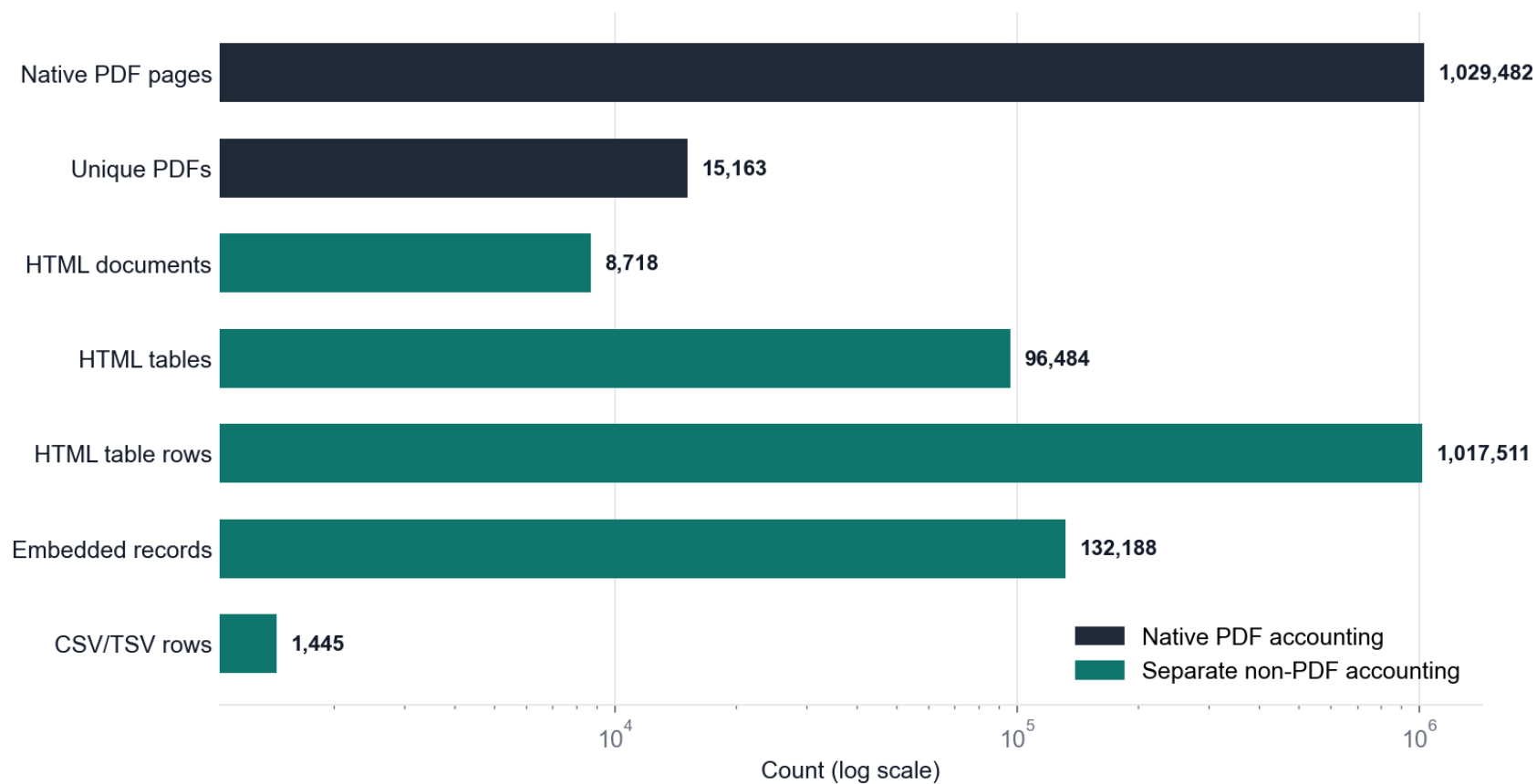
What the corpus looks like

Scale, discovery, side visibility, and geography are shown separately because their units differ.

Corpus scale and source mix

Corpus scale and source mix

Large source volume, with analytical usefulness varying by layer



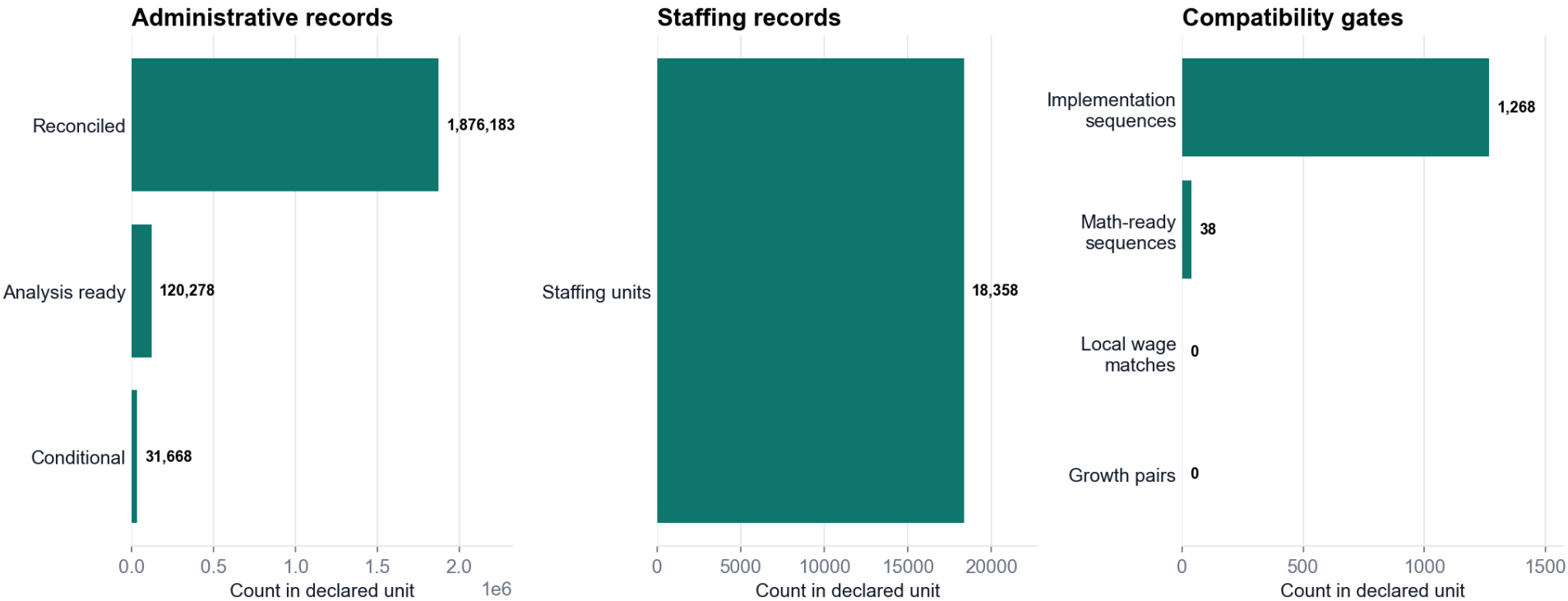
Unit: declared source or content unit • seven plotted counts • Evidence: audited corpus scale
Text-page equivalent: 650,482 (reported separately; never added to native pages).

Native PDF pages and the separate non-PDF text-page equivalent are never added together.

Evidence pipeline and responsible attrition

Evidence pipeline and analytical readiness

Filtering preserves incompatible and unresolved evidence instead of forcing it into estimates



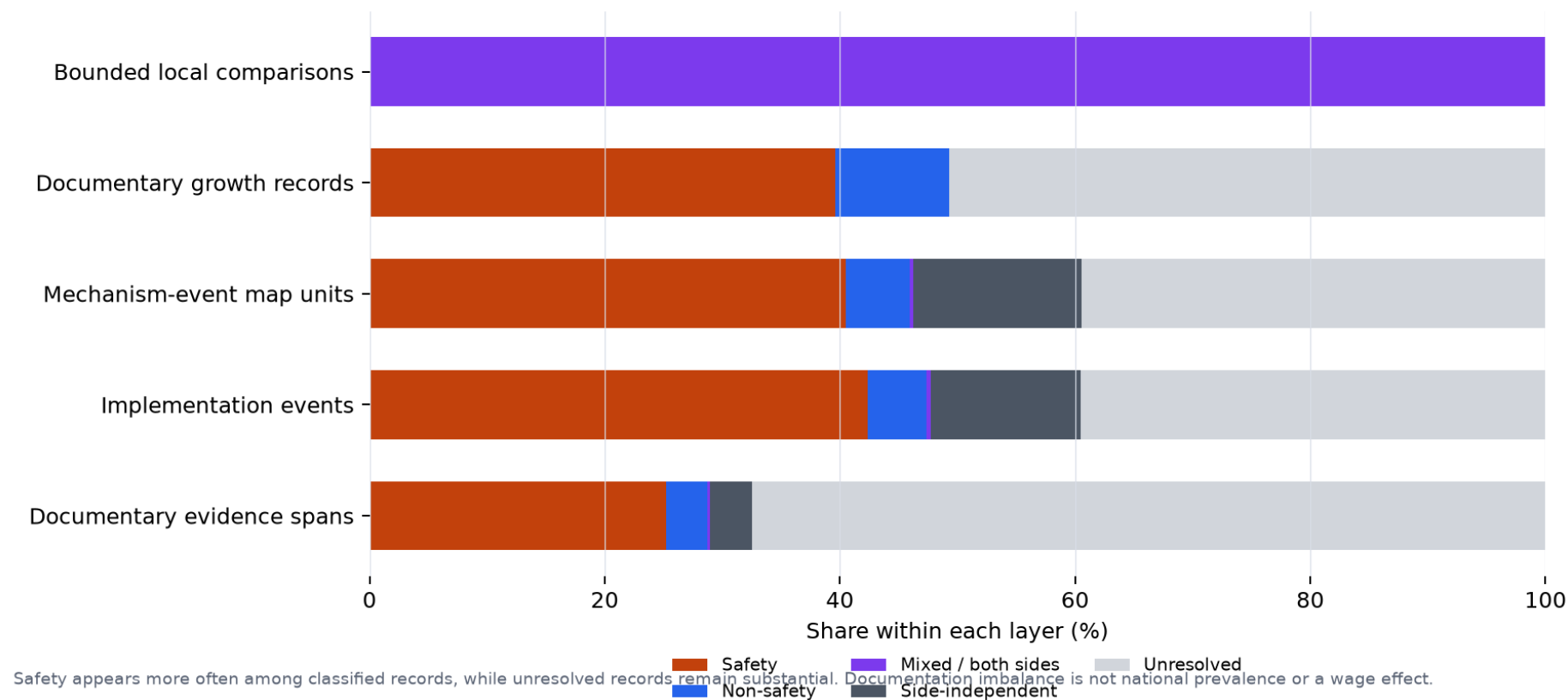
Unit: stage-specific unit • denominators shown in bounded data • Evidence: strict readiness gates
Panels use different units; they are not one continuous attrition denominator.

Filtering, deduplication, holds, incompatibility, and failures are distinct outcomes, not one undifferentiated loss.

Safety and non-safety visibility across layers

Safety and non-safety visibility differs across evidence layers

Each bar has its own analytical unit and denominator; bars are not added together

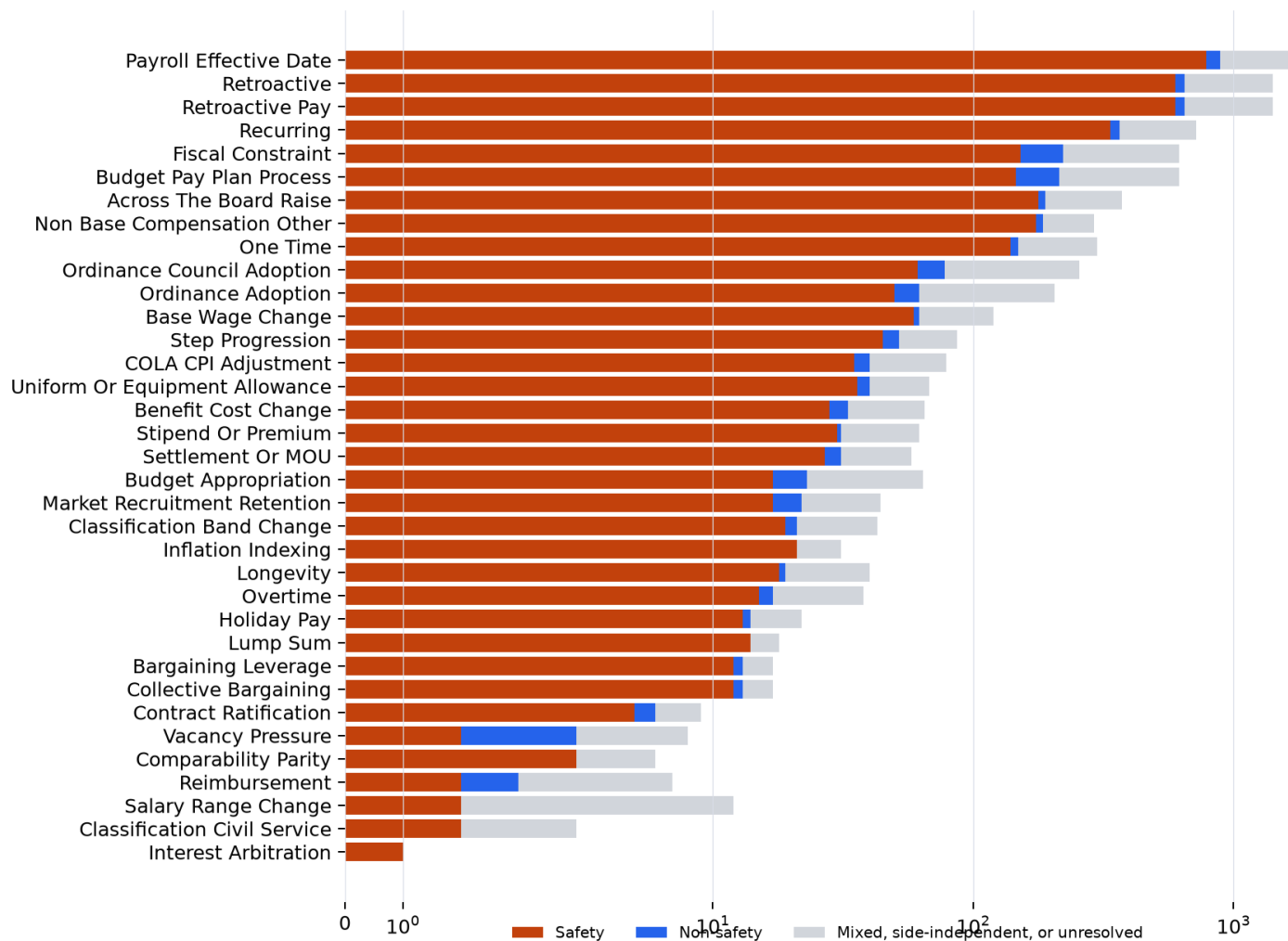


Safety appears more often among classified evidence, while unresolved shares remain large in several layers. Unlike units are not combined.

Side composition by compensation category

Side composition by compensation category

Exact counts; horizontal scale is symmetric-log so sparse categories remain visible



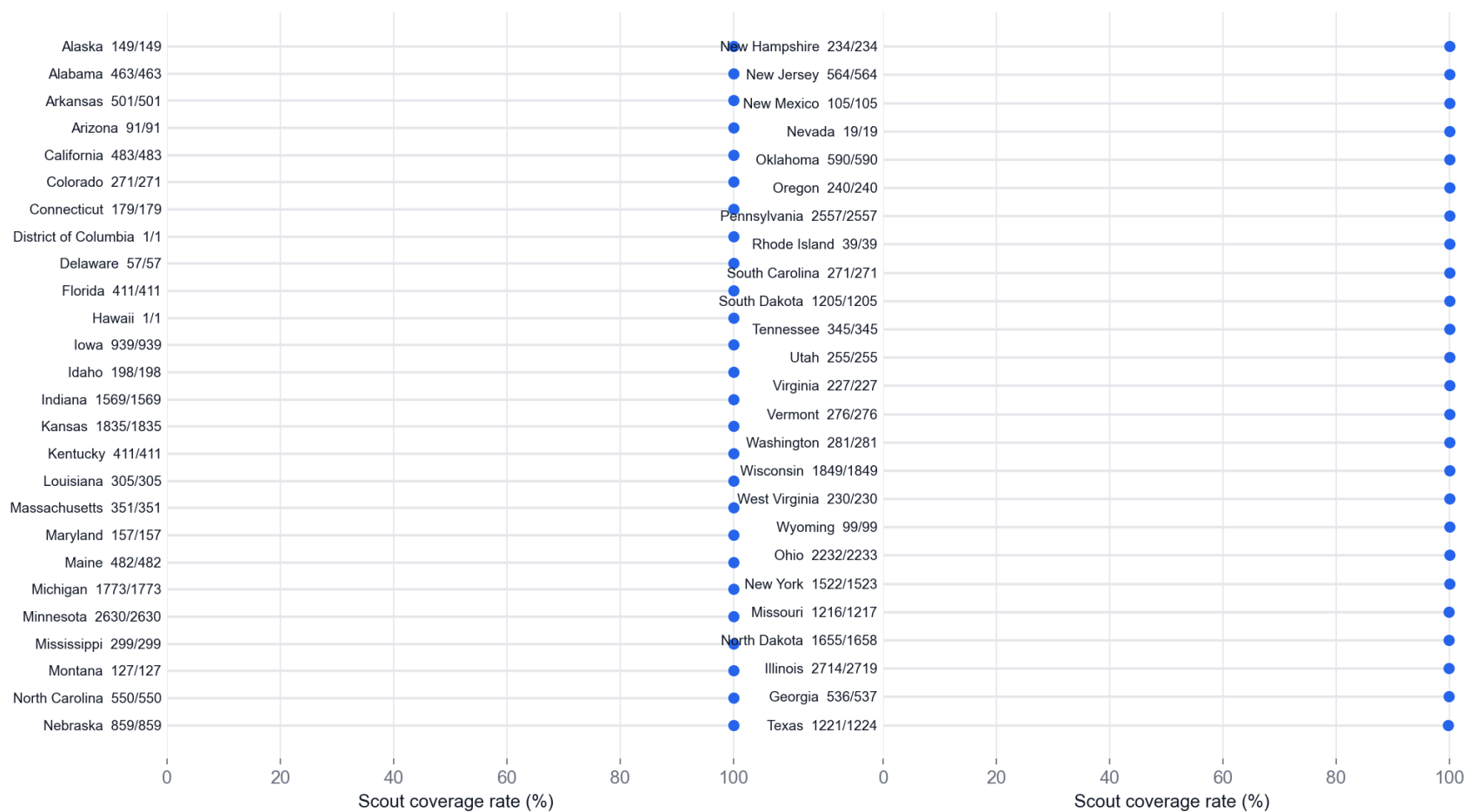
The side-classified denominator excludes mixed, side-independent, and unresolved records. Category counts use the corrected map unit.

The safety-heavy pattern is descriptive of retained and classified evidence; it is not national prevalence or causality.

Geographic scouting coverage

Municipality scout coverage by state

Coverage is the share of eligible municipalities reached—not raw source volume



Unit: eligible municipality • 51 states / district • Evidence: coverage metric

Coverage = scout-covered municipalities ÷ eligible municipality universe; not evidence quality or prevalence.

Scout coverage measures where discovery ran, not the quality or direction of evidence found.

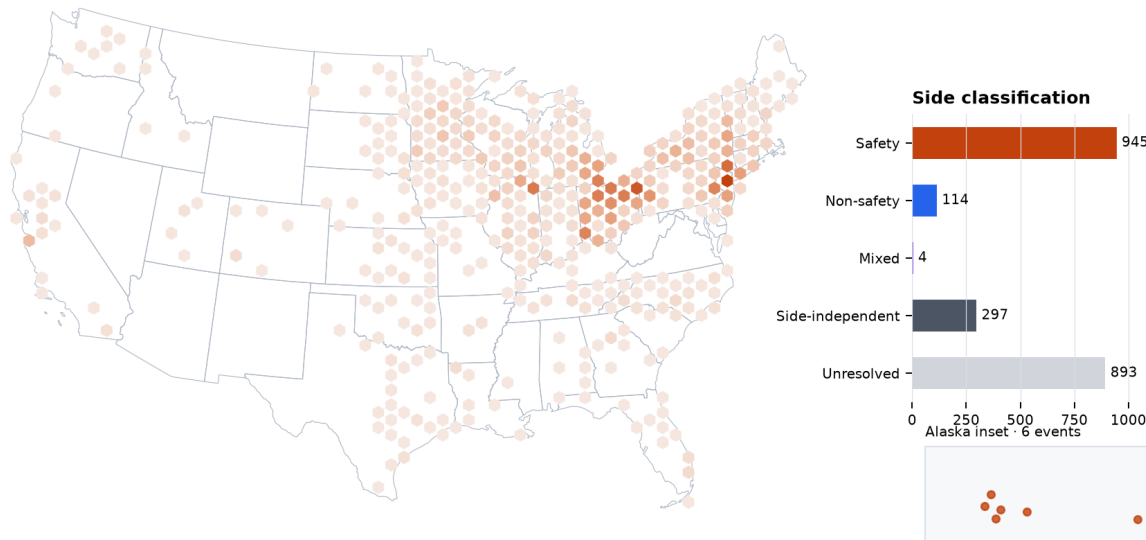
PART III

Compensation mechanisms and related claims

Eight integrated profiles explain the pay channel, side pattern, map evidence, claim boundary, and principal limitation.

Retroactivity and payroll timing

2,253 distinct implementation events · fixed lower-48 extent · Alaska retained separately



Map unit: municipality × compensation cycle × compensation mechanism × side implementation event. Fixed 50 km EPSG:5070 grid.

What this means

A compensation term whose effective date reaches back before the final approval or payment date.

How it can affect pay

It can push compensation upward by making a later settlement effective for earlier work. It converts the earlier effective period into an owed payment, often delivered as back pay. It determines when an approved increase starts reaching paychecks and prevents adoption dates from being mistaken for payment.

Side pattern

Safety 945 · Non-safety 114 · Other or unresolved 1,194

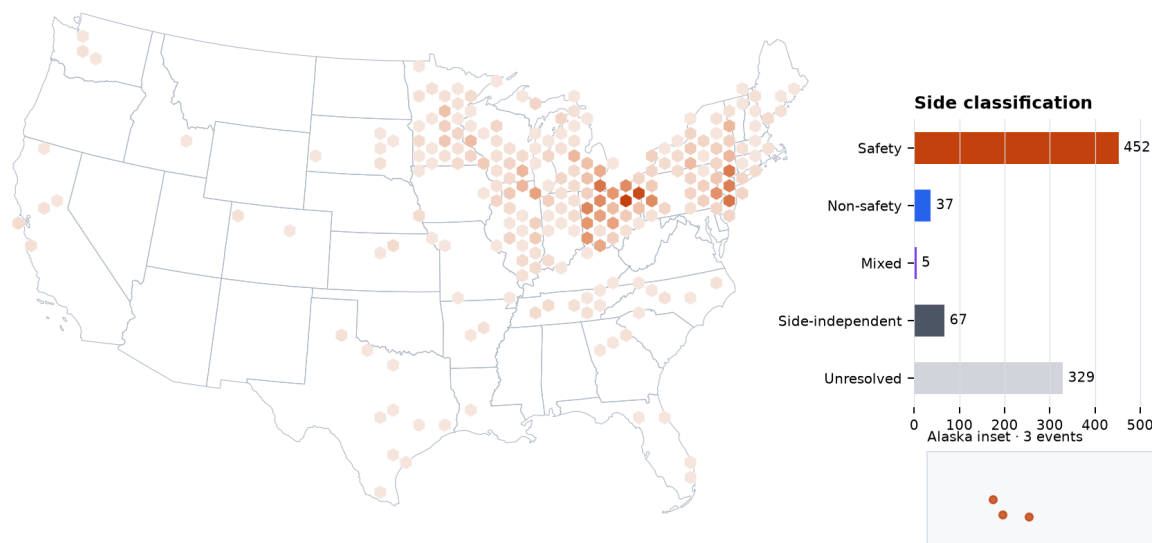
Related final claim · CLAIM-E

Reviewed records show that retroactive effective dates and implementation clauses can convert some bargaining delays into payable increases or back pay; adoption alone is not payment, and only retained evidence with a paid stage is described as paid.

A compensation term whose effective date reaches back before the final approval or payment date. It can push compensation upward by making a later settlement effective for earlier work. It converts the earlier effective period into an owed payment, often delivered as back pay. It determines when an approved increase starts reaching paychecks and prevents adoption dates from being mistaken for payment. Among side-classified events, safety appears more often (945 versus 114) in the retained evidence. That pattern matters because it shows which side is more visible in the documents I could classify, but safety records were also easier to identify and more common in the corpus. It does not establish national prevalence or a wage effect.

Non-base and one-time compensation

890 distinct implementation events · fixed lower-48 extent · Alaska retained separately



Map unit: municipality × compensation cycle × compensation mechanism × side implementation event. Fixed 50 km EPSG:5070 grid.

What this means

Compensation outside regular base wages that does not fit a more specific category.

How it can affect pay

It adds pay outside the base rate, so a base-wage comparison can miss part of compensation. It raises pay for specific duties, qualifications, shifts, or working conditions. It increases cash compensation or reimburses required job costs without changing base pay. It can substantially increase earnings beyond base pay when staffing or scheduling produces extra hours. It adds premium compensation for holiday work or recognized holidays. It adds scheduled pay as service accumulates and can steepen wage growth for long-tenured workers. It can change total compensation or take-home value even when base wages do not move. It offsets work expenses but should not be treated as recurring wages. It raises current compensation without permanently increasing the base schedule. It increases compensation once but does not by itself raise future base wages.

Side pattern

Safety 452 · Non-safety 37 · Other or unresolved 401

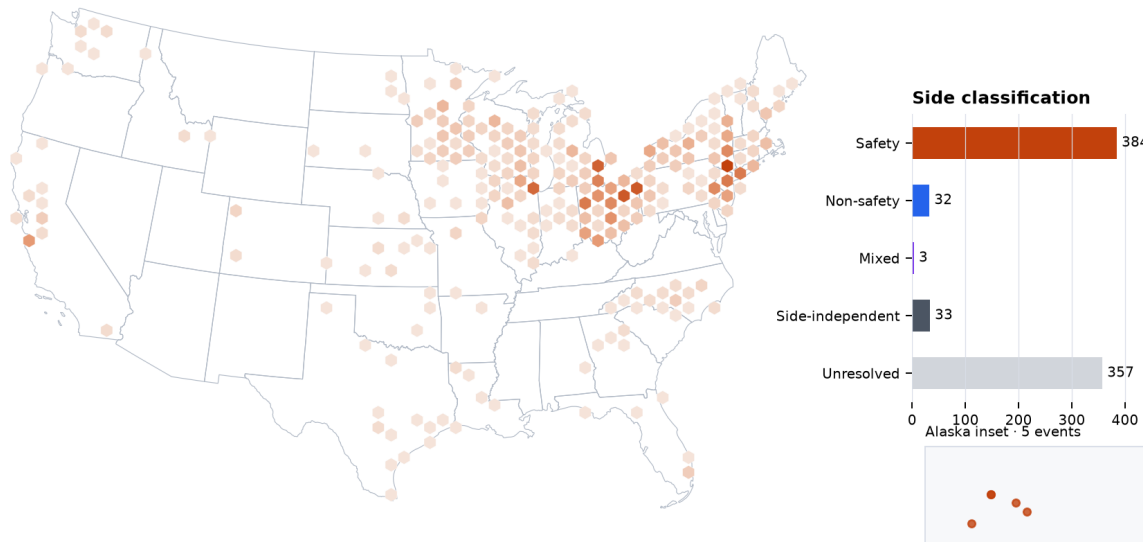
Related final claim · CLAIM-C

Reviewed safety-compensation records contain structured non-base channels—including overtime, holiday, longevity, premium, stipend, and allowance provisions—that can raise compensation beyond base pay; no complete total-compensation sum is estimated.

Compensation outside regular base wages that does not fit a more specific category. It adds pay outside the base rate, so a base-wage comparison can miss part of compensation. It raises pay for specific duties, qualifications, shifts, or working conditions. It increases cash compensation or reimburses required job costs without changing base pay. It can substantially increase earnings beyond base pay when staffing or scheduling produces extra hours. It adds premium compensation for holiday work or recognized holidays. It adds scheduled pay as service accumulates and can steepen wage growth for long-tenured workers. It can change total compensation or take-home value even when base wages do not move. It offsets work expenses but should not be treated as recurring wages. It raises current compensation without permanently increasing the base schedule. It increases compensation once but does not by itself raise future base wages. Among side-classified events, safety appears more often (452 versus 37) in the retained evidence. That pattern matters because it shows which side is more visible in the documents I could...

Scheduled base-wage growth

809 distinct implementation events · fixed lower-48 extent · Alaska retained separately



Map unit: municipality × compensation cycle × compensation mechanism × side implementation event. Fixed 50 km EPSG:5070 grid.

What this means

The same general increase applied across covered jobs or classifications.

How it can affect pay

It raises covered rates together, but does not necessarily change the gap between safety and non-safety jobs. It directly changes recurring salary or wage rates. It can generate repeated wage growth as employees move through scheduled steps. It can preserve purchasing power by linking raises to inflation or a stated percentage. It can trigger recurring adjustments as inflation changes. It raises future compensation repeatedly because the amount remains in the pay structure.

Side pattern

Safety 384 · Non-safety 32 · Other or unresolved 393

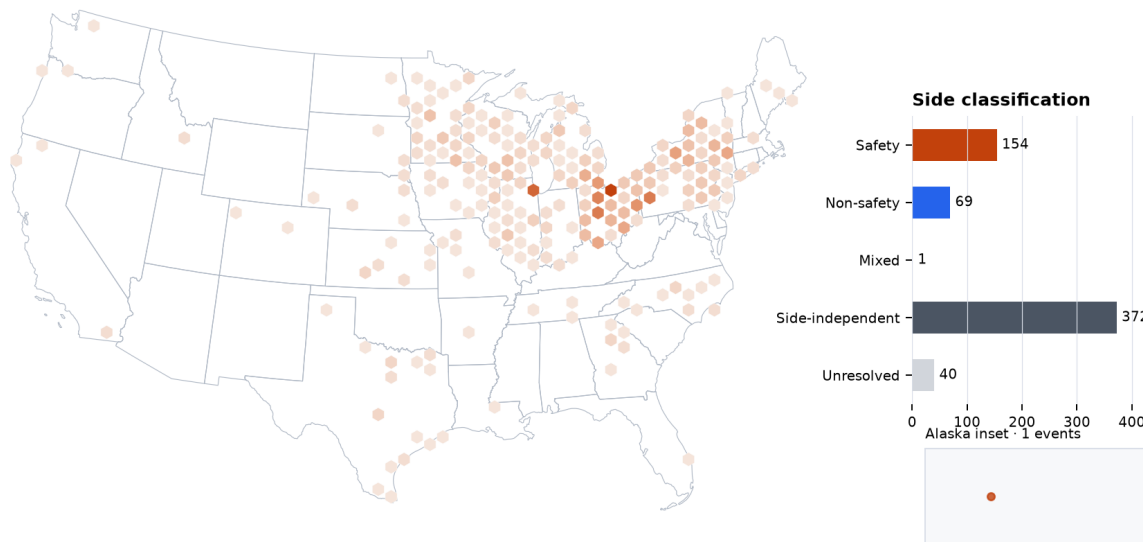
Related final claim · CLAIM-B

In reviewed unit-cycle records, step, seniority, rank, and COLA provisions create recurring scheduled wage-growth paths; step progression leans safety in the available sample, while across-board results are mixed and COLA cells are sparse.

The same general increase applied across covered jobs or classifications. It raises covered rates together, but does not necessarily change the gap between safety and non-safety jobs. It directly changes recurring salary or wage rates. It can generate repeated wage growth as employees move through scheduled steps. It can preserve purchasing power by linking raises to inflation or a stated percentage. It can trigger recurring adjustments as inflation changes. It raises future compensation repeatedly because the amount remains in the pay structure. Among side-classified events, safety appears more often (384 versus 32) in the retained evidence. That pattern matters because it shows which side is more visible in the documents I could classify, but safety records were also easier to identify and more common in the corpus. It does not establish national prevalence or a wage effect.

Budgets, pay plans, and fiscal constraint

636 distinct implementation events · fixed lower-48 extent · Alaska retained separately



Map unit: municipality × compensation cycle × compensation mechanism × side implementation event. Fixed 50 km EPSG:5070 grid.

What this means

A budget or pay-plan process that formally sets or revises compensation schedules.

How it can affect pay

It can authorize, sequence, or constrain the schedules through which wages become operative. It can hold pay down, delay implementation, or redirect increases toward narrower groups or one-time payments. It supplies the legal or financial authority for compensation to be paid.

Side pattern

Safety 154 · Non-safety 69 · Other or unresolved 413

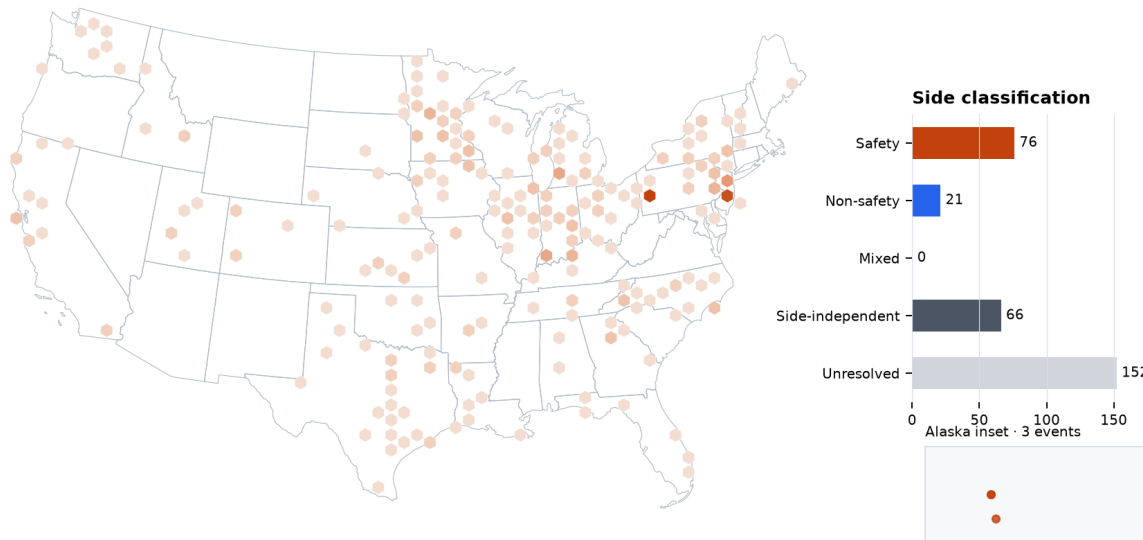
Related final claim · CLAIM-F

Reviewed ordinances, budgets, pay plans, and classification records show how compensation proposals can be formalized or constrained by fiscal institutions; these records do not establish that every proposal was implemented or paid.

A budget or pay-plan process that formally sets or revises compensation schedules. It can authorize, sequence, or constrain the schedules through which wages become operative. It can hold pay down, delay implementation, or redirect increases toward narrower groups or one-time payments. It supplies the legal or financial authority for compensation to be paid. Among side-classified events, safety appears more often (154 versus 69) in the retained evidence. That pattern matters because it shows which side is more visible in the documents I could classify, but safety records were also easier to identify and more common in the corpus. It does not establish national prevalence or a wage effect.

Council and ordinance adoption

315 distinct implementation events · fixed lower-48 extent · Alaska retained separately



Map unit: municipality × compensation cycle × compensation mechanism × side implementation event. Fixed 50 km EPSG:5070 grid.

What this means

A city council or ordinance action that formally authorizes compensation terms.

How it can affect pay

It turns a proposal or agreement into an authorized municipal action, although authorization is not proof of payment. It formalizes compensation terms, but the retained evidence must still show implementation or payment separately.

Side pattern

Safety 76 · Non-safety 21 · Other or unresolved 218

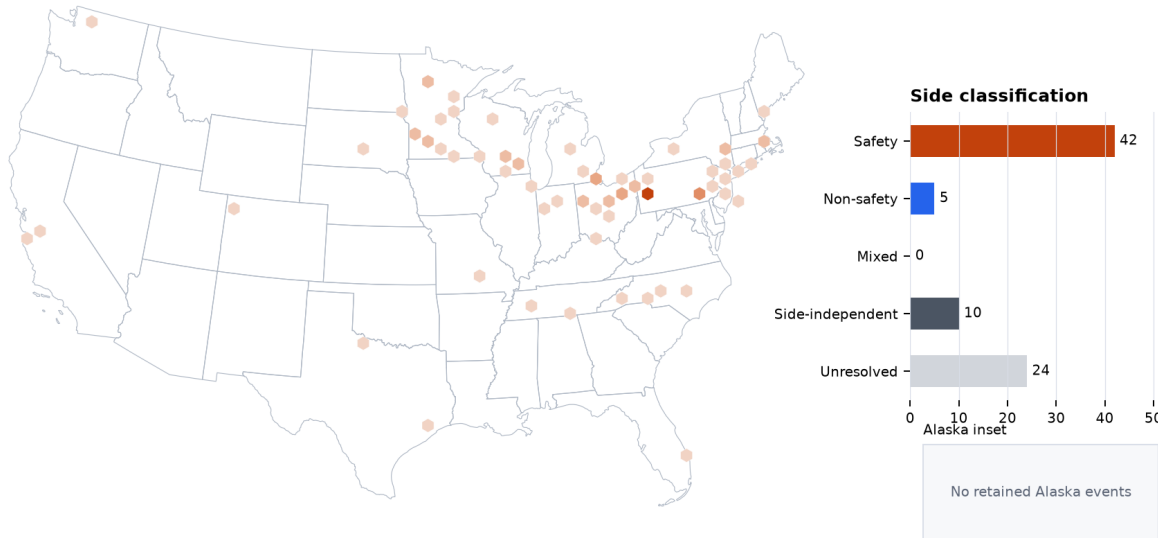
Related final claim · CLAIM-F

Reviewed ordinances, budgets, pay plans, and classification records show how compensation proposals can be formalized or constrained by fiscal institutions; these records do not establish that every proposal was implemented or paid.

A city council or ordinance action that formally authorizes compensation terms. It turns a proposal or agreement into an authorized municipal action, although authorization is not proof of payment. It formalizes compensation terms, but the retained evidence must still show implementation or payment separately. Among side-classified events, safety appears more often (76 versus 21) in the retained evidence. That pattern matters because it shows which side is more visible in the documents I could classify, but safety records were also easier to identify and more common in the corpus. It does not establish national prevalence or a wage effect.

Formal bargaining and impasse resolution

81 distinct implementation events · fixed lower-48 extent · Alaska retained separately



Map unit: municipality × compensation cycle × compensation mechanism × side implementation event. Fixed 50 km EPSG:5070 grid.

What this means

The practical power employees or unions can use to win better terms during negotiations.

How it can affect pay

It can improve negotiated terms when employees can credibly apply staffing, political, legal, or job-action pressure. It provides the formal channel through which wage schedules and non-base terms are negotiated. It can implement targeted changes faster or more narrowly than a complete successor contract. It approves negotiated terms but remains distinct from their payroll-effective or paid stage. It can impose compensation terms when ordinary bargaining reaches an impasse.

Side pattern

Safety 42 · Non-safety 5 · Other or unresolved 34

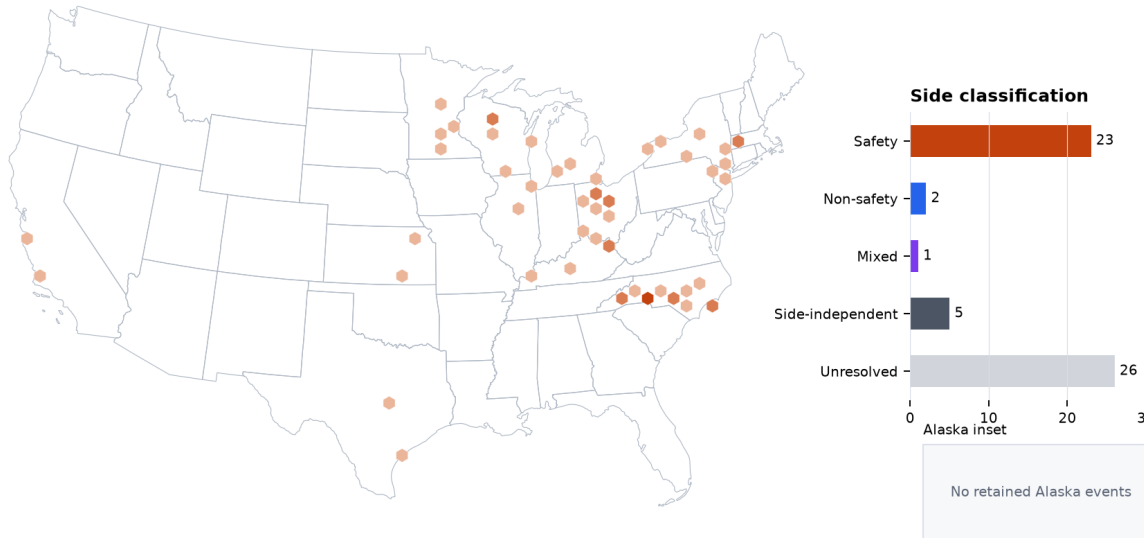
Related final claim · CLAIM-A

In the reviewed corpus, formal bargaining, arbitration, and factfinding provide recurring institutional channels that can set or preserve safety-compensation terms; the evidence establishes mechanism operation, not an average wage effect.

The practical power employees or unions can use to win better terms during negotiations. It can improve negotiated terms when employees can credibly apply staffing, political, legal, or job-action pressure. It provides the formal channel through which wage schedules and non-base terms are negotiated. It can implement targeted changes faster or more narrowly than a complete successor contract. It approves negotiated terms but remains distinct from their payroll-effective or paid stage. It can impose compensation terms when ordinary bargaining reaches an impasse. Among side-classified events, safety appears more often (42 versus 5) in the retained evidence. That pattern matters because it shows which side is more visible in the documents I could classify, but safety records were also easier to identify and more common in the corpus. It does not establish national prevalence or a wage effect.

Classification and pay structure

57 distinct implementation events · fixed lower-48 extent · Alaska retained separately



Map unit: municipality × compensation cycle × compensation mechanism × side implementation event. Fixed 50 km EPSG:5070 grid.

What this means

A change to a job classification, pay grade, range, or band.

How it can affect pay

It can raise or lower attainable pay by moving a job into a different grade or range. It changes the floor, ceiling, or spread of possible pay rather than necessarily changing observed earnings immediately. It governs placement and progression in ways that can structure long-run pay paths.

Side pattern

Safety 23 · Non-safety 2 · Other or unresolved 32

Related final claim · CLAIM-B

In reviewed unit-cycle records, step, seniority, rank, and COLA provisions create recurring scheduled wage-growth paths; step progression leans safety in the available sample, while across-board results are mixed and COLA cells are sparse.

A change to a job classification, pay grade, range, or band. It can raise or lower attainable pay by moving a job into a different grade or range. It changes the floor, ceiling, or spread of possible pay rather than necessarily changing observed earnings immediately. It governs placement and progression in ways that can structure long-run pay paths. Among side-classified events, safety appears more often (23 versus 2) in the retained evidence. That pattern matters because it shows which side is more visible in the documents I could classify, but safety records were also easier to identify and more common in the corpus. It does not establish national prevalence or a wage effect.

Staffing, market, and comparability pressure

56 distinct implementation events · fixed lower-48 extent · Alaska retained separately



Map unit: municipality × compensation cycle × compensation mechanism × side implementation event. Fixed 50 km EPSG:5070 grid.

What this means

A pay action justified by labor-market competition, hiring, or retention pressure.

How it can affect pay

It can produce targeted raises or premiums when employers struggle to hire or keep workers. It can motivate targeted raises, premiums, or retention measures when positions are hard to fill. It can pull pay toward a reference job, bargaining unit, or neighboring jurisdiction.

Side pattern

Safety 22 · Non-safety 7 · Other or unresolved 27

Related final claim · CLAIM-A

In the reviewed corpus, formal bargaining, arbitration, and factfinding provide recurring institutional channels that can set or preserve safety-compensation terms; the evidence establishes mechanism operation, not an average wage effect.

A pay action justified by labor-market competition, hiring, or retention pressure. It can produce targeted raises or premiums when employers struggle to hire or keep workers. It can motivate targeted raises, premiums, or retention measures when positions are hard to fill. It can pull pay toward a reference job, bargaining unit, or neighboring jurisdiction. Among side-classified events, safety appears more often (22 versus 7) in the retained evidence. That pattern matters because it shows which side is more visible in the documents I could classify, but safety records were also easier to identify and more common in the corpus. It does not establish national prevalence or a wage effect.

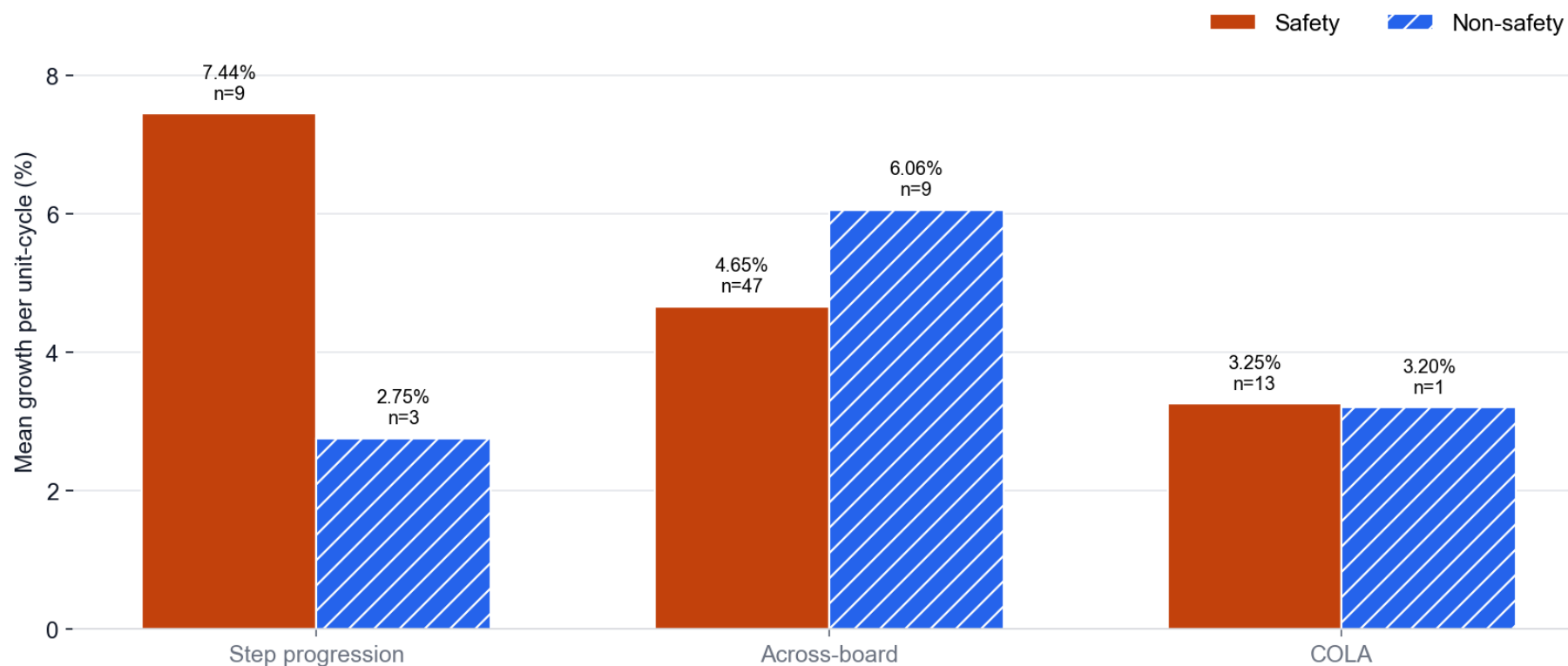
Cross-mechanism findings

Growth, staffing, implementation, local comparisons, and counterexamples are shown without turning local evidence into national claims.

Documentary wage-growth evidence

Documentary wage-growth evidence is mechanism-specific

Step progression leans safety; across-board results are mixed; COLA evidence is sparse



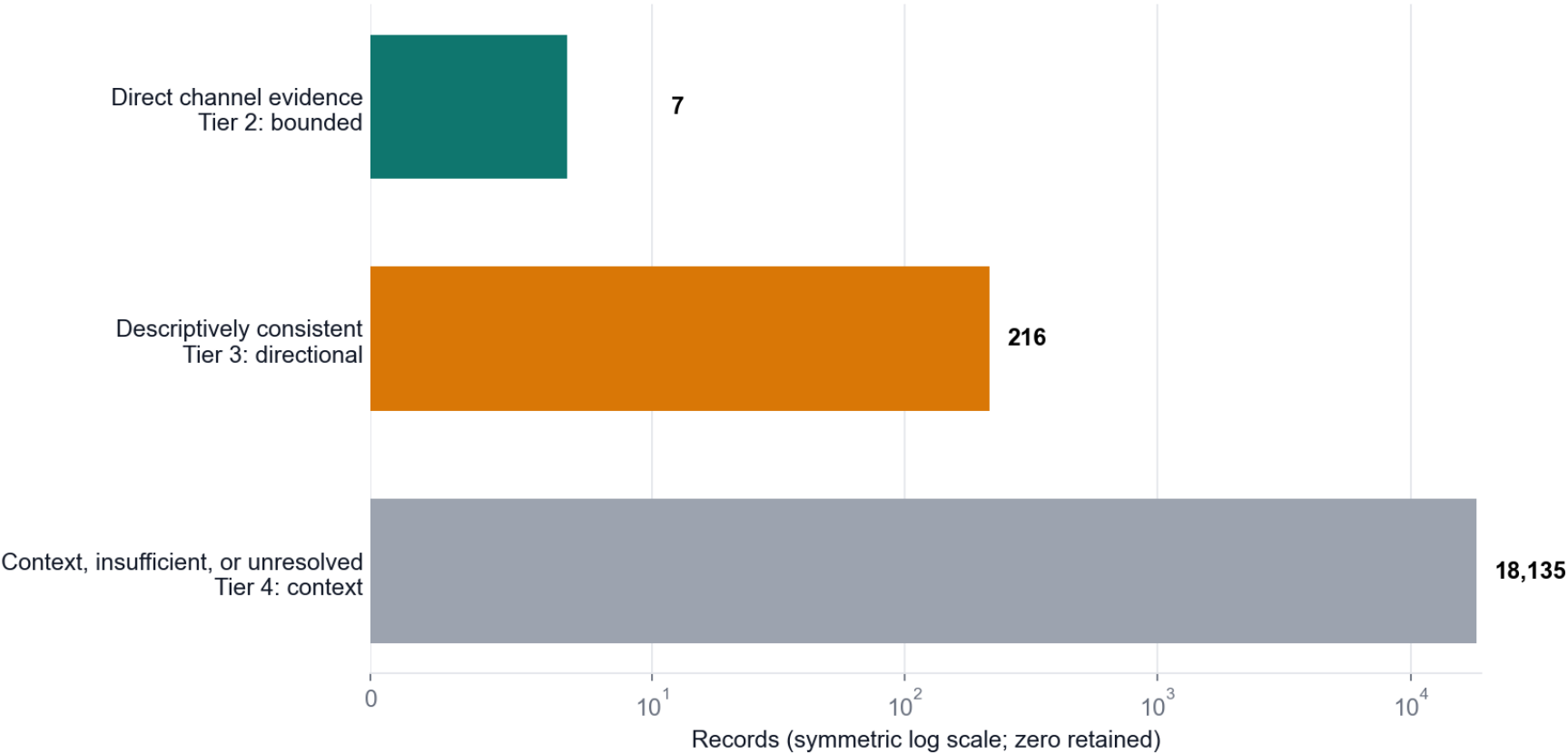
Unit: documentary unit-cycle growth record • cell n shown above each bar • Evidence: Tier 1–2 numeric summary; Tier 3 directional context
Weighted means are sample-specific; COLA cells are sparse and no uniform safety advantage is inferred.

Step progression leans safety; across-board results are mixed; COLA evidence is sparse; no uniform safety advantage is established.

Staffing-channel evidence

Staffing-channel evidence by tier

A small reviewed subset directly or descriptively fits the proposed channels



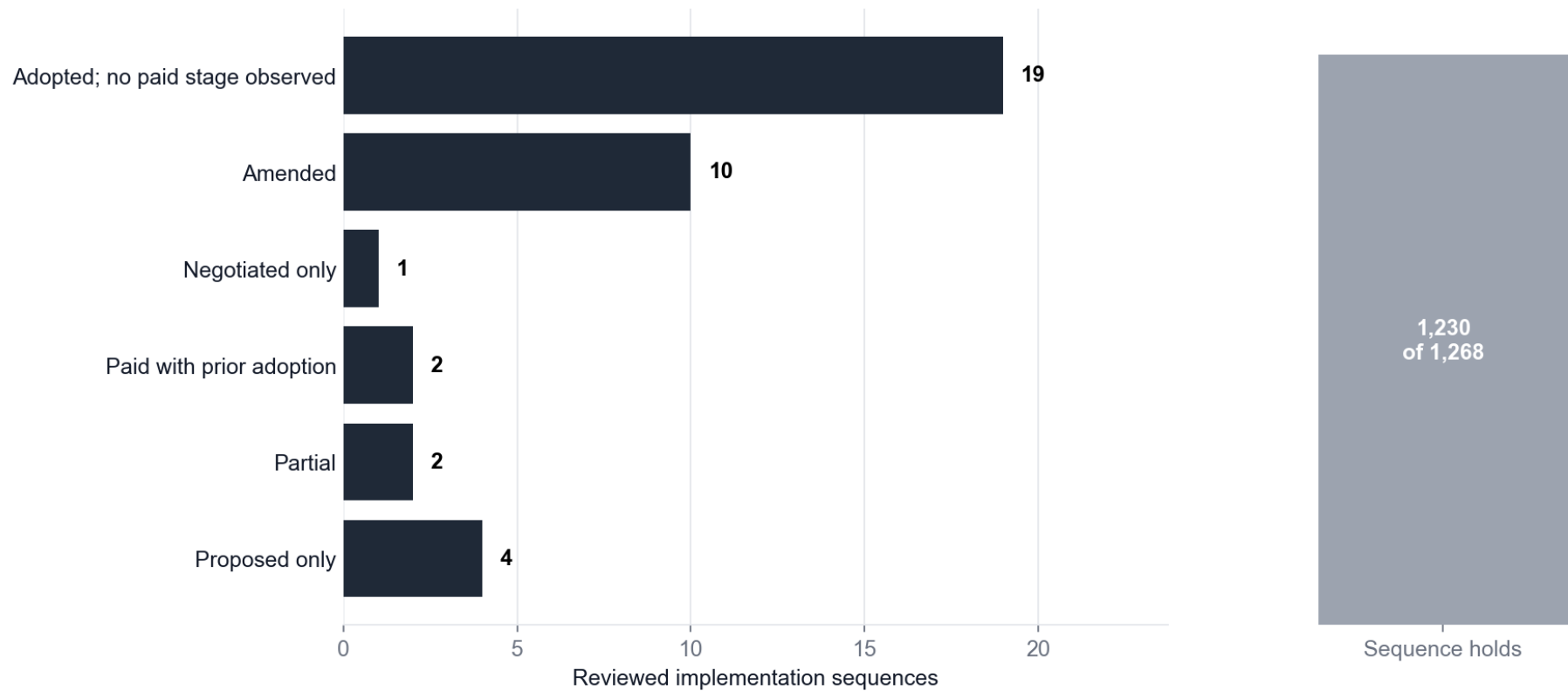
Unit: reviewed staffing record • n = 18,358 • Evidence: Tier 2 / Tier 3 / context
Direct means source-explicit channel evidence—not an estimated causal effect.

Seven direct channel records and 216 descriptively consistent records support a noncausal staffing-pressure interpretation.

Implementation lifecycle

Implementation evidence distinguishes adoption from payment

Adoption, amendment, implementation, and payment remain distinct stages



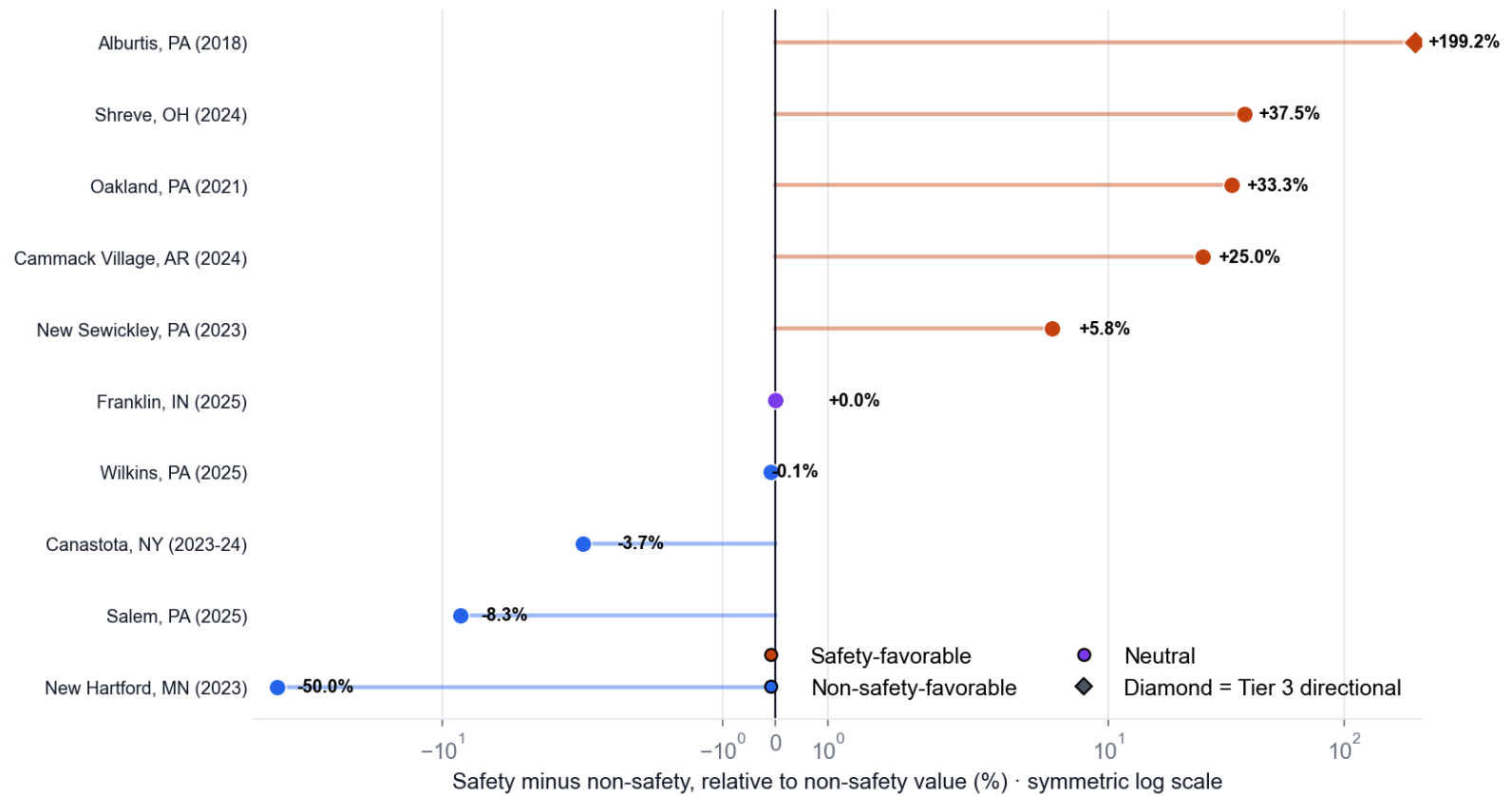
Unit: deduplicated implementation sequence • 38 reviewed; 1,230 held; denominator = 1,268 • Evidence: strict reviewed sequences
“No paid stage observed in retained evidence” is not “never paid.”

Adoption remains distinct from payment. “No paid stage observed” is not the same as “never paid.”

Ten bounded local comparisons

Bounded local wage comparisons point in both directions

Five favor safety, four favor non-safety, and one is neutral



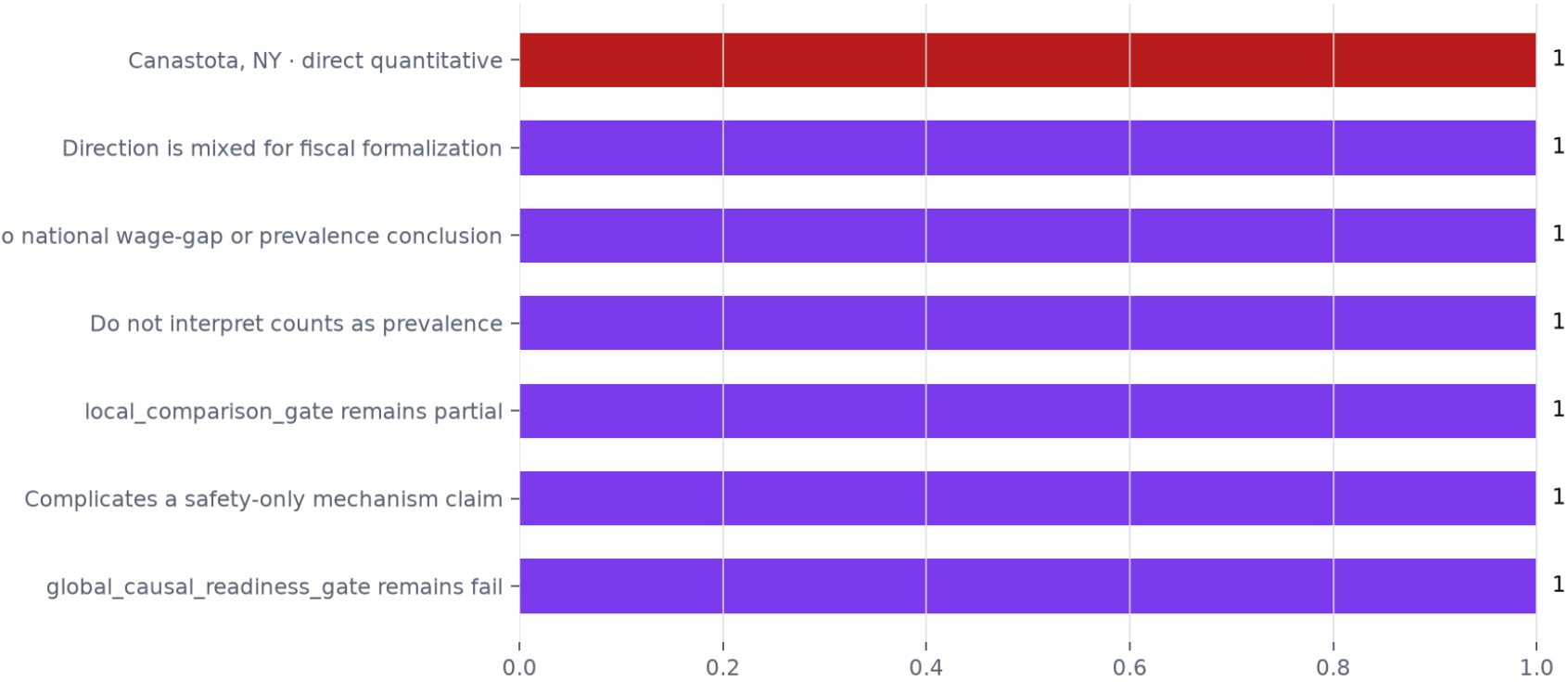
Unit: bounded same-municipality comparison • $n = 10$; 5 safety-favorable, 4 non-safety-favorable, 1 neutral • Evidence: nine Tier 2; one Tier 3
 Role, period, and pay-basis caveats remain. These local examples are not a national estimate.

Five favor safety, four favor non-safety, and one is neutral. These are local examples, not a national estimate.

Counterexamples and countervailing evidence

Seven counterexamples remain part of the result

One direct quantitative example and six qualitative or mechanism-bounding records



Counterexamples define the boundary of defensible claims; they are not decorative limitations.

One direct quantitative counterexample and six qualitative or mechanism-bounding records remain part of the result.

PART V

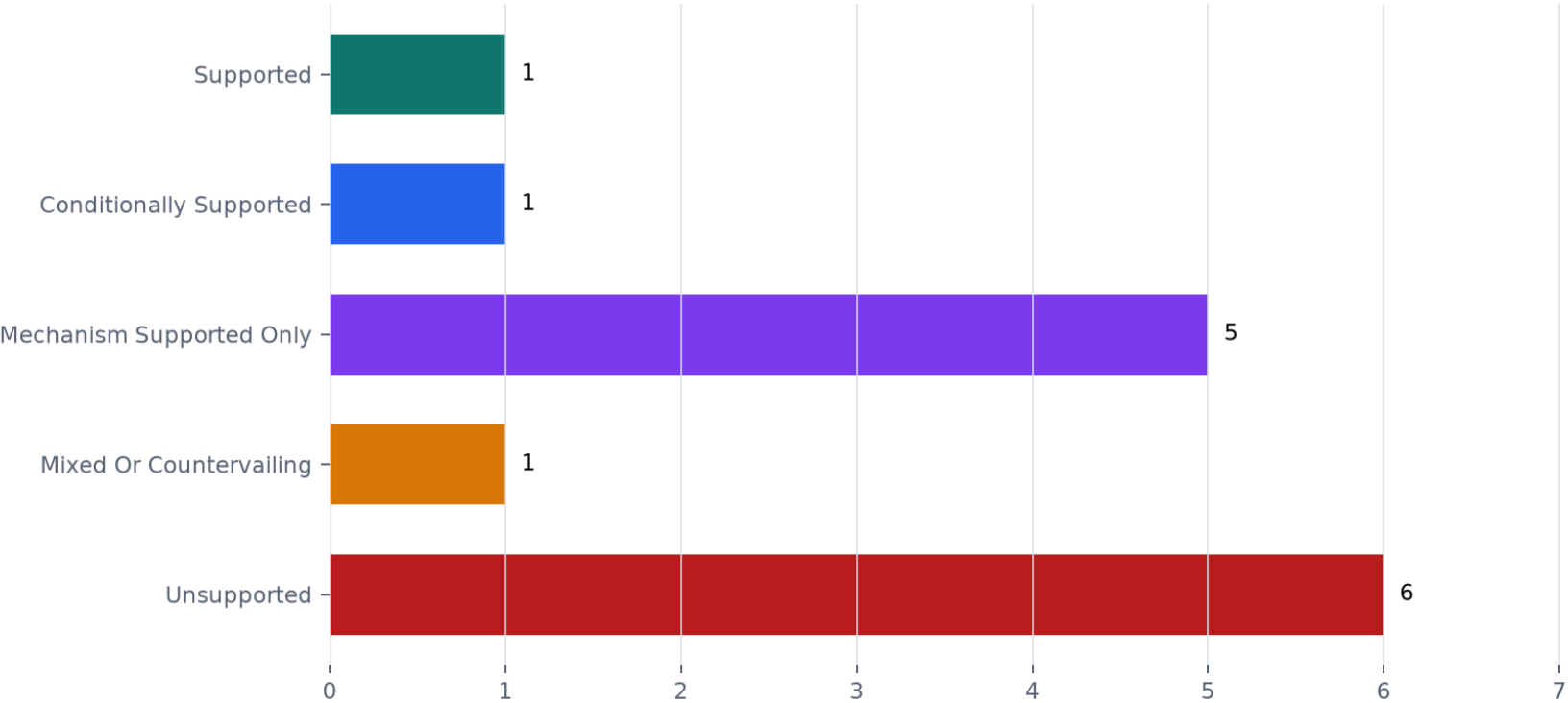
What the claims support

Every final class and wording boundary is preserved. Mechanism profiles replace duplicative cards where possible; compact cards preserve the full technical record.

Final claim overview

Final claim classes

All 14 claims retain their final adjudicated class



One supported · one conditional · five mechanism-supported only · one mixed · six unsupported

The final universe remains one supported, one conditional, five mechanism-supported, one mixed, and six unsupported.

Claim evidence matrix

Claim evidence matrix

Complete adjudicated wording appears on the immediately following claim-card pages; cell numbers are actual linked-record counts

CLAIM-A — Formal bargaining and impasse resolution	6	0	1	0	0
CLAIM-B — Scheduled wage-growth mechanisms	7	6	423	0	0
CLAIM-C — Non-base compensation channels	2	0	1	0	0
CLAIM-D — Staffing and labor-market pressure	4	7	217	0	0
CLAIM-E — Retroactivity and implementation timing	40	0	1	0	0
CLAIM-F — Fiscal institutions and formal adoption	44	0	1	0	0
CLAIM-G — Mixed and countervailing evidence	11	15	424	7	0
CLAIM-H — A bounded compensation-mechanism account	52	23	677	7	0
UNSUP-01 — No national safety wage-gap estimate	0	0	0	0	0
UNSUP-02 — No national mechanism-prevalence estimate	0	0	0	0	0
UNSUP-03 — No causal-effect estimate	0	0	0	0	0
UNSUP-04 — No regression-based claim	0	0	0	0	0
UNSUP-05 — No fixed safety growth advantage	0	0	0	0	0
UNSUP-06 — Documentation is not causal identification	0	0	0	0	0
	Strict evidence	Bounded evidence	Directional evidence	Counter-examples	Conflicts

Color intensity uses $\log(1 + \text{count})$. Counts are evidence links, not equally weighted proof. Unresolved conflicts remain excluded from clean support totals.

Full claim wording is shown; cell numbers are actual linked-record counts and color intensity uses a declared log transformation.

Claim boundaries — complete wording

Formal bargaining and impasse resolution

CLAIM-A · mechanism supported only · report supporting finding

What I can responsibly say

In the reviewed corpus, formal bargaining, arbitration, and factfinding provide recurring institutional channels that can set or preserve safety-compensation terms; the evidence establishes mechanism operation, not an average wage effect.

Strict evidence

In the reviewed corpus, formal bargaining, arbitration, and factfinding provide recurring institutional channels that can set or preserve safety-compensation terms.

Broader bounded evidence

In the reviewed corpus, formal bargaining, arbitration, and factfinding provide recurring institutional channels that can set or preserve safety-compensation terms; the evidence establishes mechanism operation, not an average wage effect.

What I cannot responsibly say

Formal bargaining causes safety wages to rise faster nationwide.

Main uncertainty: unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain

Scheduled wage-growth mechanisms

CLAIM-B · conditionally supported · report major section

What I can responsibly say

In reviewed unit-cycle records, step, seniority, rank, and COLA provisions create recurring scheduled wage-growth paths; step progression leans safety in the available sample, while across-board results are mixed and COLA cells are sparse.

Strict evidence

The reviewed documentary corpus contains recurring step, seniority, rank, and COLA structures; comparative growth implications remain sample-specific.

Broader bounded evidence

In reviewed unit-cycle records, step, seniority, rank, and COLA provisions create recurring scheduled wage-growth paths; step progression leans safety in the available sample, while across-board results are mixed and COLA cells are sparse.

What I cannot responsibly say

Safety wages generally grow faster nationwide because of step progression or COLAs.

Main uncertainty: unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain

Claim boundaries — complete wording

Non-base compensation channels

CLAIM-C · mechanism supported only · report supporting finding

What I can responsibly say

Reviewed safety-compensation records contain structured non-base channels—including overtime, holiday, longevity, premium, stipend, and allowance provisions—that can raise compensation beyond base pay; no complete total-compensation sum is estimated.

Strict evidence

Reviewed safety-compensation records contain structured non-base channels—including overtime, holiday, longevity, premium, stipend, and allowance provisions—that can raise compensation beyond base pay.

Broader bounded evidence

Reviewed safety-compensation records contain structured non-base channels—including overtime, holiday, longevity, premium, stipend, and allowance provisions—that can raise compensation beyond base pay; no complete total-compensation sum is estimated.

What I cannot responsibly say

Safety workers receive more total compensation than non-safety workers nationwide.

Main uncertainty: unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain

Staffing and labor-market pressure

CLAIM-D · mechanism supported only · report supporting finding

What I can responsibly say

Reviewed sources explicitly use recruitment, retention, comparator, vacancy, and staffing pressure to justify schedule changes or targeted premiums in some public-safety settings; the staffing evidence is descriptive, not causal or prevalence evidence.

Strict evidence

Reviewed sources explicitly use recruitment, retention, comparator, vacancy, and staffing pressure to justify schedule changes or targeted premiums in some public-safety settings.

Broader bounded evidence

Reviewed sources explicitly use recruitment, retention, comparator, vacancy, and staffing pressure to justify schedule changes or targeted premiums in some public-safety settings; the staffing evidence is descriptive, not causal or prevalence evidence.

What I cannot responsibly say

Staffing shortages cause higher safety wages.

Main uncertainty: unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain

Claim boundaries — complete wording

Retroactivity and implementation timing

CLAIM-E · mechanism supported only · report supporting finding

What I can responsibly say

Reviewed records show that retroactive effective dates and implementation clauses can convert some bargaining delays into payable increases or back pay; adoption alone is not payment, and only retained evidence with a paid stage is described as paid.

Strict evidence

Reviewed records show that retroactive effective dates and implementation clauses can convert some bargaining delays into payable increases or back pay.

Broader bounded evidence

Reviewed records show that retroactive effective dates and implementation clauses can convert some bargaining delays into payable increases or back pay; adoption alone is not payment, and only retained evidence with a paid stage is described as paid.

What I cannot responsibly say

Every adopted or retroactive provision was implemented and paid.

Main uncertainty: unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain

Fiscal institutions and formal adoption

CLAIM-F · mechanism supported only · report supporting finding

What I can responsibly say

Reviewed ordinances, budgets, pay plans, and classification records show how compensation proposals can be formalized or constrained by fiscal institutions; these records do not establish that every proposal was implemented or paid.

Strict evidence

Reviewed ordinances, budgets, pay plans, and classification records show how compensation proposals can be formalized or constrained by fiscal institutions.

Broader bounded evidence

Reviewed ordinances, budgets, pay plans, and classification records show how compensation proposals can be formalized or constrained by fiscal institutions; these records do not establish that every proposal was implemented or paid.

What I cannot responsibly say

Budget and council action uniformly increase compensation.

Main uncertainty: unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain

Claim boundaries — complete wording

Mixed and countervailing evidence

CLAIM-G · mixed or countervailing · report counterexample section

What I can responsibly say

Both safety and non-safety units use bargaining, steps, COLAs, and pay plans. The available evidence is consistent with a reinforcing safety-pressure bundle in some settings, but local and growth results are mixed and do not establish a uniform safety advantage.

Strict evidence

Safety and non-safety units both use formal compensation-growth mechanisms; the comparative pattern in reviewed records is mixed.

Broader bounded evidence

Both safety and non-safety units use bargaining, steps, COLAs, and pay plans. The available evidence is consistent with a reinforcing safety-pressure bundle in some settings, but local and growth results are mixed and do not establish a uniform safety advantage.

What I cannot responsibly say

Non-safety compensation lacks institutional growth mechanisms, or safety always has the advantage.

Main uncertainty: unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain

A bounded compensation-mechanism account

CLAIM-H · supported · report core argument

What I can responsibly say

The reviewed corpus supports a bounded account of mechanisms and directional pressures affecting safety compensation more strongly than it supports any global wage-gap, prevalence, or causal-effect estimate.

Strict evidence

The reviewed corpus supports a bounded account of mechanisms and directional pressures affecting safety compensation more strongly than it supports any global wage-gap, prevalence, or causal-effect estimate.

Broader bounded evidence

The reviewed corpus supports a bounded account of mechanisms and directional pressures affecting safety compensation more strongly than it supports any global wage-gap, prevalence, or causal-effect estimate.

What I cannot responsibly say

The corpus estimates a national safety wage gap or causal effect.

Main uncertainty: unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain

Claim boundaries — complete wording

No national safety wage-gap estimate

UNSUP-01 · unsupported · report limitations

What I can responsibly say

A national safety wage-gap estimate is not supported by the current corpus.

Strict evidence

A national safety wage-gap estimate is not supported by the current corpus.

Broader bounded evidence

A national safety wage-gap estimate is not supported by the current corpus.

What I cannot responsibly say

The corpus provides a national safety wage-gap estimate.

Main uncertainty: unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain

No national mechanism-prevalence estimate

UNSUP-02 · unsupported · report limitations

What I can responsibly say

A national prevalence estimate for any compensation mechanism is not supported by the current corpus.

Strict evidence

A national prevalence estimate for any compensation mechanism is not supported by the current corpus.

Broader bounded evidence

A national prevalence estimate for any compensation mechanism is not supported by the current corpus.

What I cannot responsibly say

The corpus estimates national mechanism prevalence.

Main uncertainty: unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain

Claim boundaries — complete wording

No causal-effect estimate

UNSUP-03 · unsupported · report limitations

What I can responsibly say

A causal effect estimate for any compensation mechanism is not supported by the current design.

Strict evidence

A causal effect estimate for any compensation mechanism is not supported by the current design.

Broader bounded evidence

A causal effect estimate for any compensation mechanism is not supported by the current design.

What I cannot responsibly say

The corpus identifies causal effects of compensation mechanisms.

Main uncertainty: unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain

No regression-based claim

UNSUP-04 · unsupported · appendix only

What I can responsibly say

No regression-based claim is supported because no regression-ready design passed the gate and no regression was run.

Strict evidence

No regression-based claim is supported because no regression-ready design passed the gate and no regression was run.

Broader bounded evidence

No regression-based claim is supported because no regression-ready design passed the gate and no regression was run.

What I cannot responsibly say

Regression results establish the proposed relationship.

Main uncertainty: unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain

Claim boundaries — complete wording

No fixed safety growth advantage

UNSUP-05 · unsupported · report limitations

What I can responsibly say

A precise claim that safety wages grow a fixed percentage faster is not supported by the available matched evidence.

Strict evidence

A precise claim that safety wages grow a fixed percentage faster is not supported by the available matched evidence.

Broader bounded evidence

A precise claim that safety wages grow a fixed percentage faster is not supported by the available matched evidence.

What I cannot responsibly say

Safety wages grow X percent faster nationally.

Main uncertainty: unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain

Documentation is not causal identification

UNSUP-06 · unsupported · report limitations

What I can responsibly say

The evidence does not establish that a documented mechanism caused an observed wage difference.

Strict evidence

The evidence does not establish that a documented mechanism caused an observed wage difference.

Broader bounded evidence

The evidence does not establish that a documented mechanism caused an observed wage difference.

What I cannot responsibly say

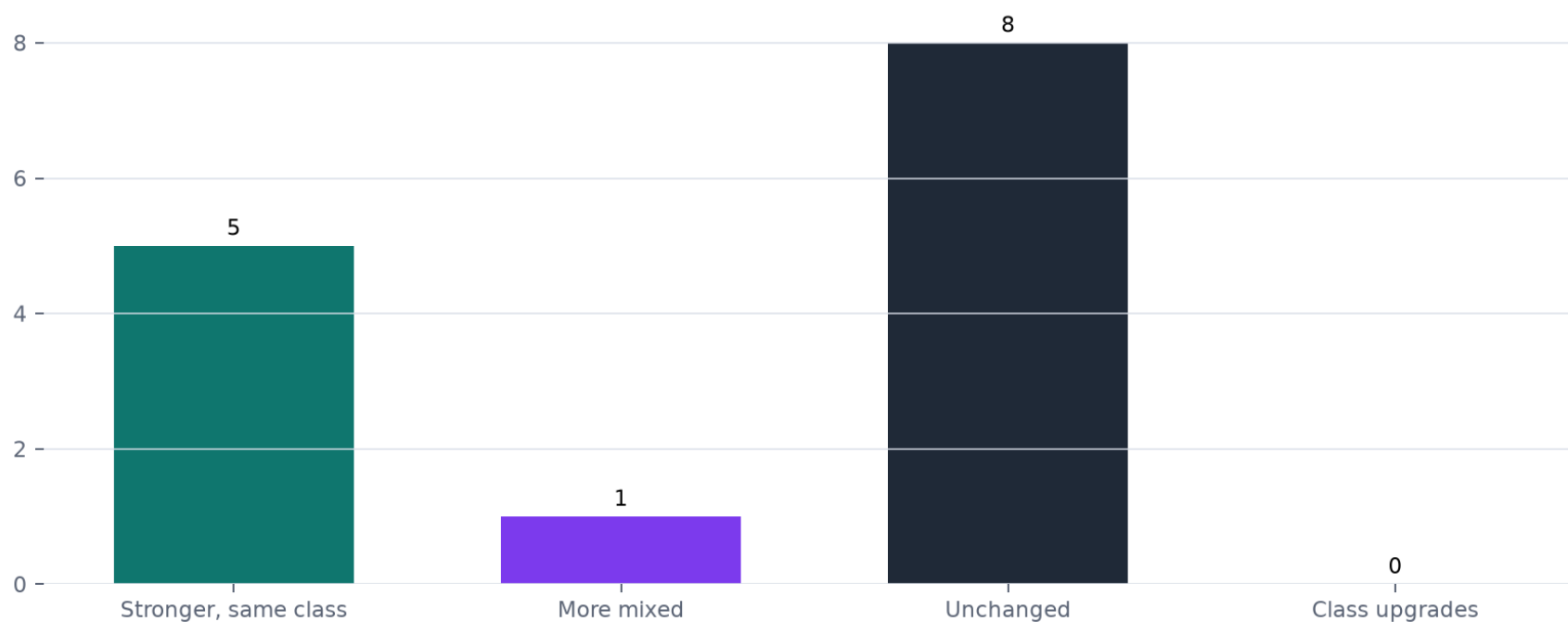
The documented mechanism caused the observed wage difference.

Main uncertainty: unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain

Strict versus broader evidence

Broader evidence changed strength, not claim classes

Strict results were preserved



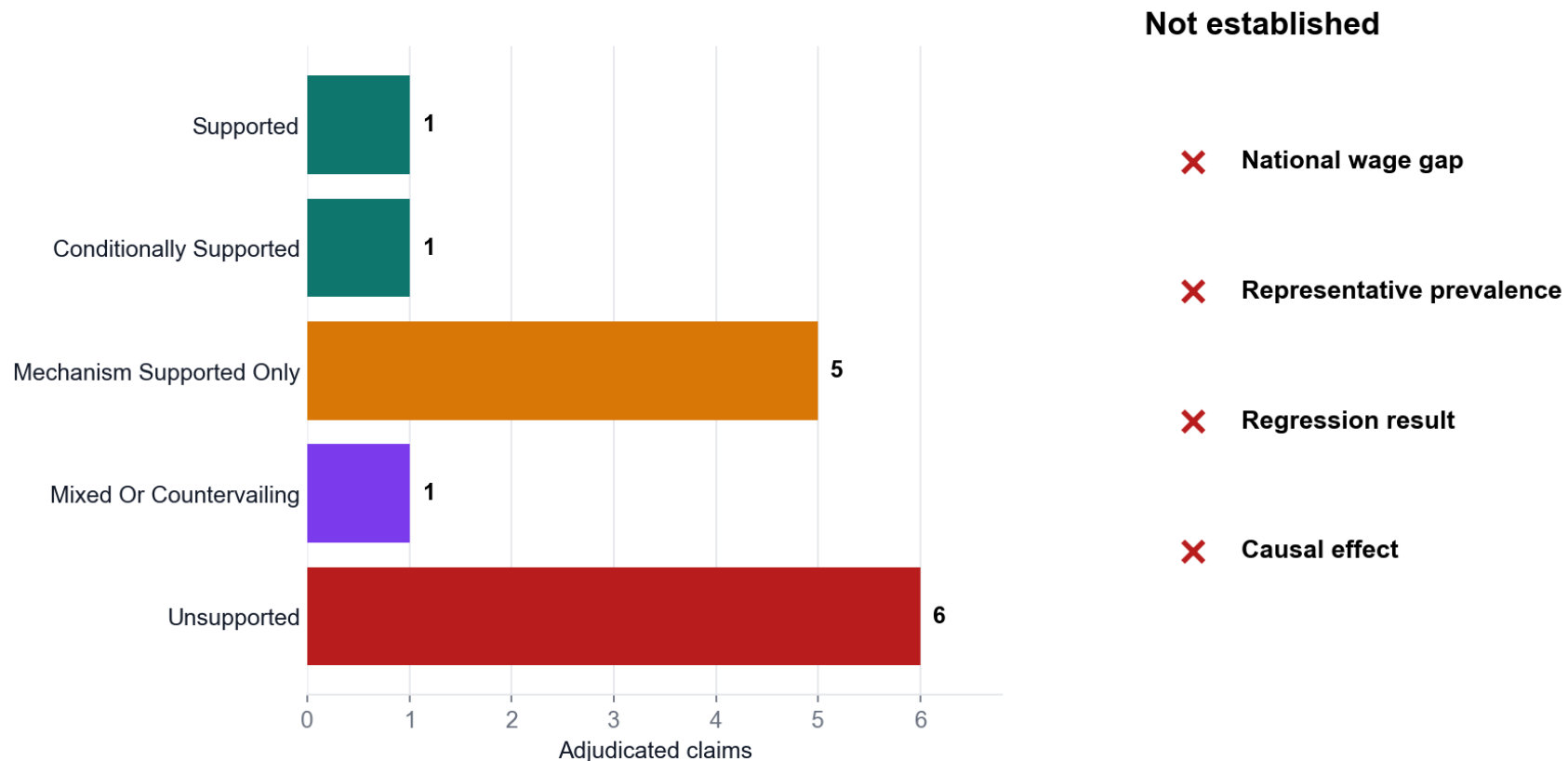
Bounded and directional evidence did not rescue unsupported claims.

Broader evidence strengthened five claims within class, made one more mixed, left eight unchanged, and upgraded none.

Final evidence boundary

What the evidence supports—and what it does not establish

The project supports bounded mechanism findings, not national or causal estimates



Unit: adjudicated report claim • n = 14 • Evidence: final strict/bounded integration
 Unsupported propositions remain explicit limitations rather than being rescued by evidence volume.

The project supports bounded mechanisms and local patterns, not a national wage gap, prevalence, regression, or causal estimate.

PART VI

What worked, what failed, and what limited the project

Each limitation is paired with what the project still accomplished and the exact boundary that remains.

Research design and scope

Research design and scope

What limited the analysis

- The corpus is not a representative national sample and has no national worker, bargaining-unit, or pay-record denominator.

What still worked

- Scouting reached 35,574 of 35,589 eligible municipalities (99.9579%), and the retained corpus contains 15,163 unique PDFs with 1,029,482 native PDF pages.

Near-complete discovery coverage is not analytical representativeness.

The corpus is not a representative national sample and has no national worker, bargaining-unit, or pay-record denominator. Scouting reached 35,574 of 35,589 eligible municipalities (99.9579%), and the retained corpus contains 15,163 unique PDFs with 1,029,482 native PDF pages. Near-complete discovery coverage is not analytical representativeness.

Discovery and hosted-search limits

Discovery and hosted-search limits

What limited the analysis

- Of 18,689 intended residual targets, 12,844 remained unsearched after repeated source-less search outcomes. Those targets are unknown possibilities, not known missing evidence.

What still worked

- Previously found candidates and verified sources remained intact, and the unsearched queue was preserved rather than treated as evidence of absence.

The failure was consistent with a service or capacity limitation, but its exact cause was not exposed.

Of 18,689 intended residual targets, 12,844 remained unsearched after repeated source-less search outcomes. Those targets are unknown possibilities, not known missing evidence. Previously found candidates and verified sources remained intact, and the unsearched queue was preserved rather than treated as evidence of absence. The failure was consistent with a service or capacity limitation, but its exact cause was not exposed.

Verification, access, and storage

Verification, access, and storage

What limited the analysis

- Access blocks, unavailable URLs, corrupt payloads, and a 30 GiB retention ceiling left 7,895 verified sources storage-held and 24,569 low-value context sources deferred.

What still worked

- The project retained 14,449 unique external payloads with hashes and source pointers and kept storage holds separate from substantive rejections.

Storage-held sources were not processed for this revision.

Access blocks, unavailable URLs, corrupt payloads, and a 30 GiB retention ceiling left 7,895 verified sources storage-held and 24,569 low-value context sources deferred. The project retained 14,449 unique external payloads with hashes and source pointers and kept storage holds separate from substantive rejections. Storage-held sources were not processed for this revision.

Document readiness

Document readiness

What limited the analysis

- PDFs, HTML, tables, CSVs, embedded data, shells, and low-text files required different paths; 118 sources were deferred for OCR and 97 needed extraction repair.

What still worked

- Of 14,703 inspected retained-source records, 14,257 were extraction-ready and 14,160 produced usable payloads.

OCR deferral means image-only documents were not silently treated as empty.

PDFs, HTML, tables, CSVs, embedded data, shells, and low-text files required different paths; 118 sources were deferred for OCR and 97 needed extraction repair. Of 14,703 inspected retained-source records, 14,257 were extraction-ready and 14,160 produced usable payloads. OCR deferral means image-only documents were not silently treated as empty.

Extraction volume and compaction

Extraction volume and compaction

What limited the analysis

- Extraction produced 5,558,770 raw field hits and 4,289,437 raw spans. Repeated tables, boilerplate, and ambiguous labels meant those were not independent facts.

What still worked

- Deterministic compaction reduced the field layer by 66.25% and preserved 1,876,183 compact administrative observations.

Large row counts measure processing volume, not evidence prevalence.

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Analytical framing and table calibration

Analytical framing and table calibration

What limited the analysis

- Early wage-table signals were not reliable enough to become a clean comparable wage panel; a five-case rendered challenge contradicted all five assisted outcomes.

What still worked

- The project did not force those signals into estimates. It expanded to staffing, implementation, documentary growth, local comparisons, and compensation mechanisms.

The broader framing improved evidence use but did not manufacture missing wage matches.

Early wage-table signals were not reliable enough to become a clean comparable wage panel; a five-case rendered challenge contradicted all five assisted outcomes. The project did not force those signals into estimates. It expanded to staffing, implementation, documentary growth, local comparisons, and compensation mechanisms. The broader framing improved evidence use but did not manufacture missing wage matches.

Rating and semantic review

Rating and semantic review

What limited the analysis

- The new external administrative layer did not receive GABRIEL scoring. Deterministic rules and bounded semantic AI review are not independent human gold coding.

What still worked

- Claim-critical review examined 1,726 unique records; 368 sampled second-pass QA records passed, and 201 conflicts remained unresolved rather than conveniently resolved.

Earlier eligible documentary evidence did use GABRIEL where available.

The new external administrative layer did not receive GABRIEL scoring. Deterministic rules and bounded semantic AI review are not independent human gold coding. Claim-critical review examined 1,726 unique records; 368 sampled second-pass QA records passed, and 201 conflicts remained unresolved rather than conveniently resolved. Earlier eligible documentary evidence did use GABRIEL where available.

Reconciliation and comparability

Reconciliation and comparability

What limited the analysis

- Side, period, role, pay basis, and compensation basis often did not align: 1,523,558 observations remained side-unresolved, with 131,124 basis holds and 266,849 conflict-sensitive observations.

What still worked

- The project preserved 18,358 staffing units, 1,268 implementation sequences, 10 bounded local comparisons, and 432 documentary growth records.

The external layer still produced zero compatible wage matches, growth pairs, vacancy rates, overtime shares, or total-compensation sums.

Side, period, role, pay basis, and compensation basis often did not align: 1,523,558 observations remained side-unresolved, with 131,124 basis holds and 266,849 conflict-sensitive observations. The project preserved 18,358 staffing units, 1,268 implementation sequences, 10 bounded local comparisons, and 432 documentary growth records. The external layer still produced zero compatible wage matches, growth pairs, vacancy rates, overtime shares, or total-compensation sums.

Safety and non-safety evidence imbalance

Safety and non-safety evidence imbalance

What limited the analysis

- Safety-specific records are more visible, while non-safety pay is often embedded in broad budgets or personnel schedules and many records remain unresolved.

What still worked

- The analysis keeps safety, non-safety, mixed, side-independent, and unresolved records separate and reports exact denominators.

Documentation absence cannot be treated as real-world absence or national prevalence.

Safety-specific records are more visible, while non-safety pay is often embedded in broad budgets or personnel schedules and many records remain unresolved. The analysis keeps safety, non-safety, mixed, side-independent, and unresolved records separate and reports exact denominators. Documentation absence cannot be treated as real-world absence or national prevalence.

Mathematical and causal design

Mathematical and causal design

What limited the analysis

- Only 3 of 16 regression-readiness gates passed. There was no defensible external matched wage panel or causal identification design.

What still worked

- No regression was run and no national wage-gap, prevalence, or causal estimate was produced; strict descriptive and mechanism findings were retained.

Not running an invalid model is a substantive quality decision.

Only 3 of 16 regression-readiness gates passed. There was no defensible external matched wage panel or causal identification design. No regression was run and no national wage-gap, prevalence, or causal estimate was produced; strict descriptive and mechanism findings were retained. Not running an invalid model is a substantive quality decision.

Locality and visual grouping

Locality and visual grouping

What limited the analysis

- Local comparisons remain local; maps show documented events, not workers, wage-effect size, or prevalence; grouping compresses source lineage.

What still worked

- All 14 claims were adjudicated, seven counterexamples remained visible, and 201 high-impact conflicts stayed outside clean totals.

Appendix tables preserve the category and claim lineage hidden by concise profiles.

Local comparisons remain local; maps show documented events, not workers, wage-effect size, or prevalence; grouping compresses source lineage. All 14 claims were adjudicated, seven counterexamples remained visible, and 201 high-impact conflicts stayed outside clean totals. Appendix tables preserve the category and claim lineage hidden by concise profiles.

Operational and infrastructure limits

Operational and infrastructure limits

What limited the analysis

- Hosted search failed, storage filled, checkpoint designs scaled poorly, some parsers hung, interfaces interrupted coordination, and deployment jobs were delayed.

What still worked

- Lane-owned files, append-only ledgers, checkpoints, quarantine, hashes, and resume-only logic preserved accepted work. Phase 0 inventoried about 119.88 GiB without deleting it.

Operational recovery protected evidence but added monitoring and reconciliation cost.

Hosted search failed, storage filled, checkpoint designs scaled poorly, some parsers hung, interfaces interrupted coordination, and deployment jobs were delayed. Lane-owned files, append-only ledgers, checkpoints, quarantine, hashes, and resume-only logic preserved accepted work. Phase 0 inventoried about 119.88 GiB without deleting it. Operational recovery protected evidence but added monitoring and reconciliation cost.

What worked, what did not, and what remains possible

What worked, what did not, and what remains possible

What worked

- 99.9579% scout coverage; more than one million native PDF pages; 14,160 usable external payloads; 1,876,183 compact observations; staffing, implementation, local-comparison, growth, and mechanism evidence.

What did not become possible

- A representative national sample; compatible external wage or growth panel; national prevalence; a regression estimate; a causal effect; a uniform safety growth advantage.

Current evidence can support focused recovery, better matched local panels, selected human review, and a bounded handoff without overstating results.

The project produced a large, traceable evidence base and bounded mechanism findings, but it did not produce a representative wage panel, national prevalence estimate, regression, or causal effect. The remaining work is targeted: improve matched local panels, resolve selected conflicts, and preserve source lineage for future research.

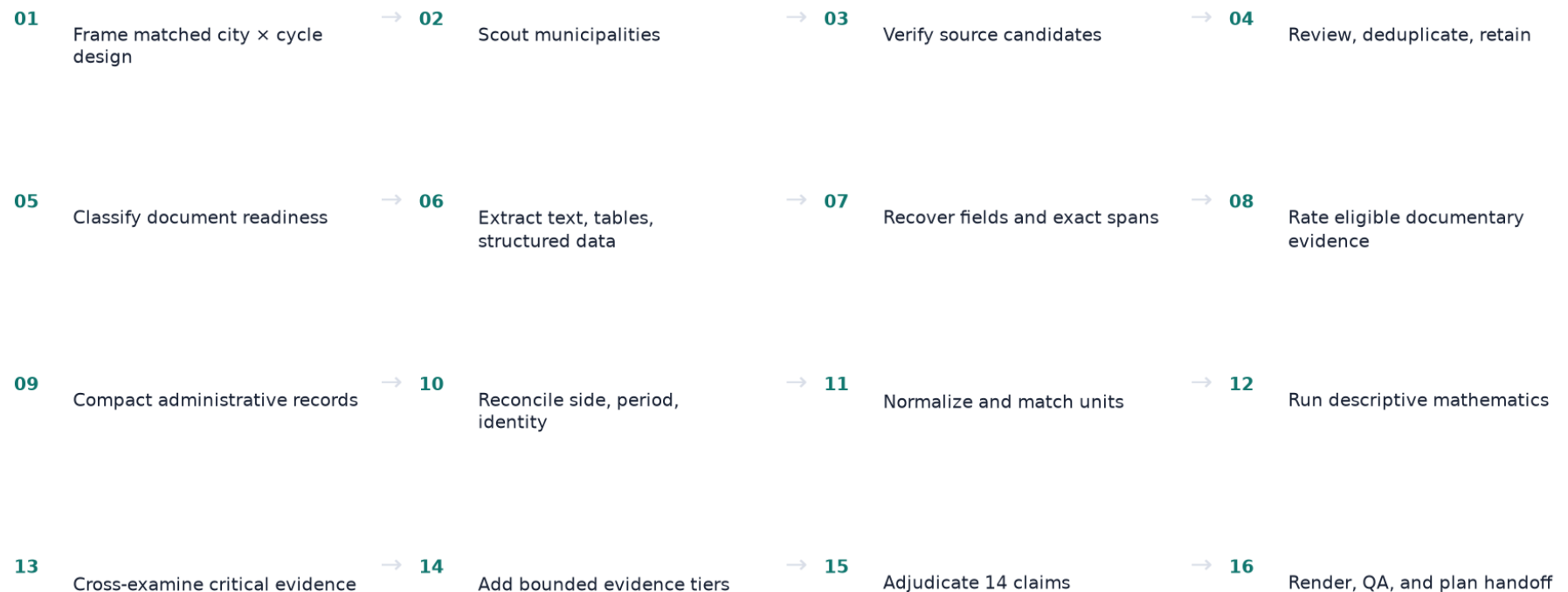
PART VII

How the project worked

The workflow combined human research direction, AI orchestration, local execution, source-grounded rating, deterministic processing, and explicit failure recovery.

Full-summer research workflow

Full-summer research workflow

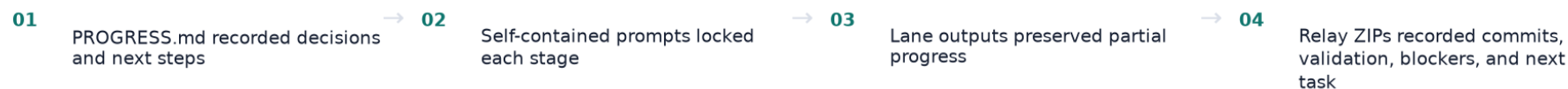


I directed the research goals and standards. ChatGPT designed the staged orchestration and evidence rules. Codex executed the local pipeline. GABRIEL rated eligible earlier documentary evidence; later administrative evidence used deterministic local processing and bounded semantic AI review.

I use this page to document full-summer research workflow and to show how the project changed as evidence, infrastructure, and scale required safer designs. The sequence preserves human direction, AI execution, accepted work, and strict claim boundaries instead of presenting the workflow as perfectly linear.

From a continuing log to relay packages

From a continuing log to relay packages



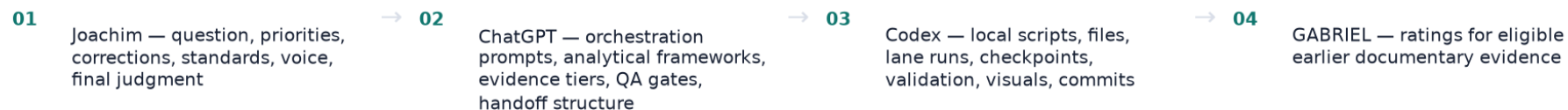
05 ChatGPT read relays and designed the next bounded prompt

I directed the research goals and standards. ChatGPT designed the staged orchestration and evidence rules. Codex executed the local pipeline. GABRIEL rated eligible earlier documentary evidence; later administrative evidence used deterministic local processing and bounded semantic AI review.

I use this page to document from a continuing log to relay packages and to show how the project changed as evidence, infrastructure, and scale required safer designs. The sequence preserves human direction, AI execution, accepted work, and strict claim boundaries instead of presenting the workflow as perfectly linear.

Human and AI division of work

Human and AI division of work



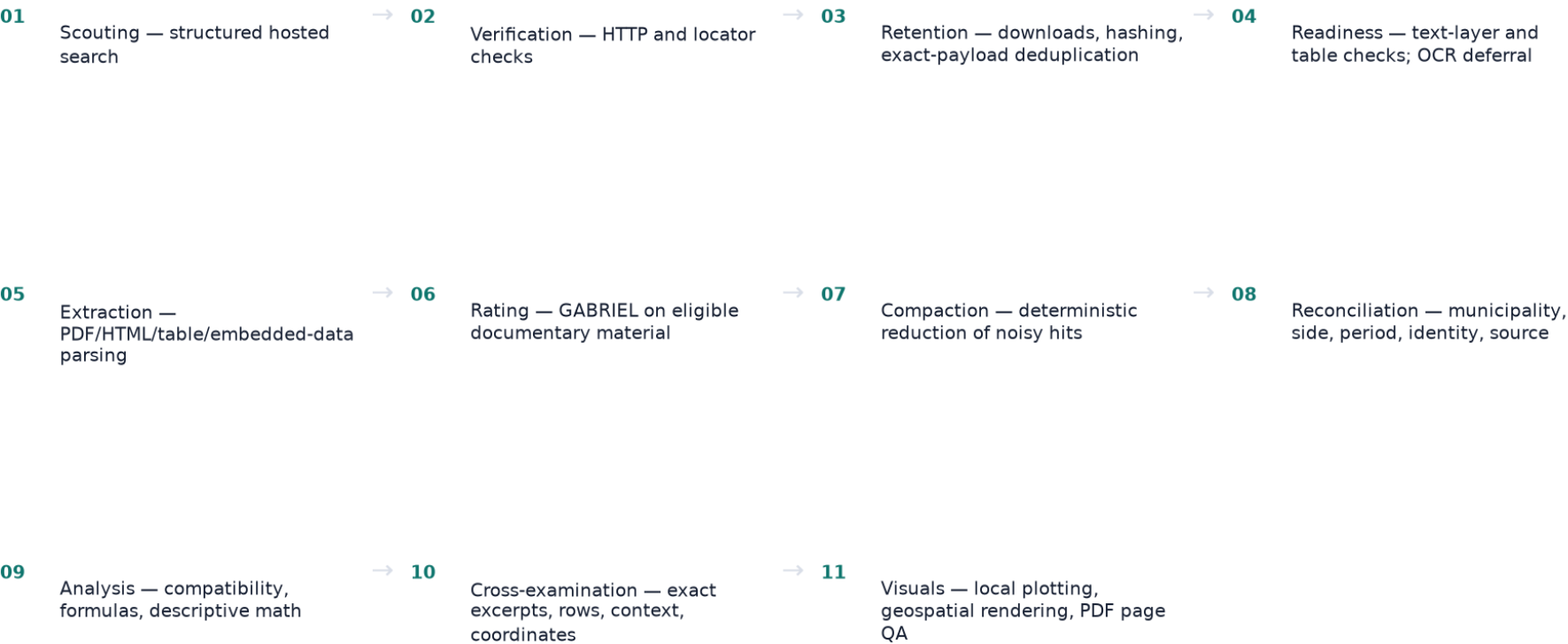
05
 Deterministic local rules —
 later administrative processing
 without new GABRIEL scoring

I directed the research goals and standards. ChatGPT designed the staged orchestration and evidence rules. Codex executed the local pipeline. GABRIEL rated eligible earlier documentary evidence; later administrative evidence used deterministic local processing and bounded semantic AI review.

I use this page to document human and ai division of work and to show how the project changed as evidence, infrastructure, and scale required safer designs. The sequence preserves human direction, AI execution, accepted work, and strict claim boundaries instead of presenting the workflow as perfectly linear.

Tools and functions by stage

Tools and functions by stage



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I use this page to document tools and functions by stage and to show how the project changed as evidence, infrastructure, and scale required safer designs. The sequence preserves human direction, AI execution, accepted work, and strict claim boundaries instead of presenting the workflow as perfectly linear.

Codex execution workflow evolved with scale

Codex execution workflow evolved with scale

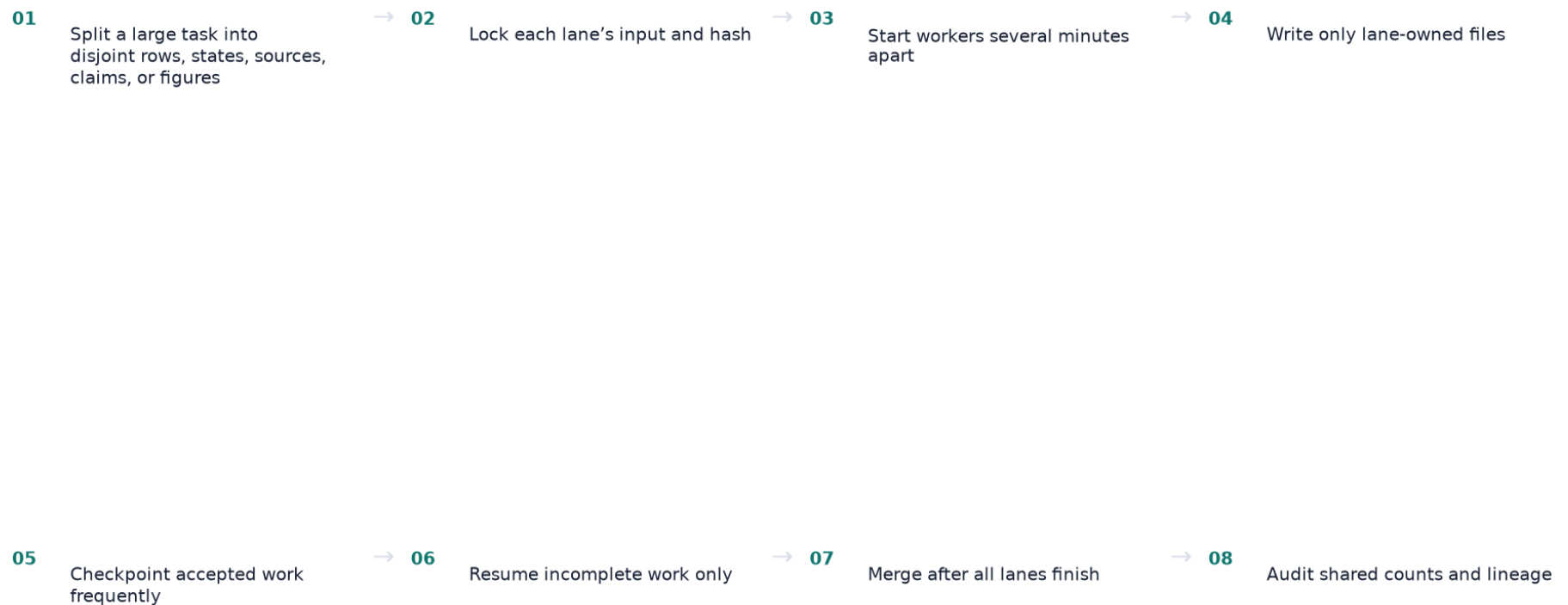


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I use this page to document codex execution workflow evolved with scale and to show how the project changed as evidence, infrastructure, and scale required safer designs. The sequence preserves human direction, AI execution, accepted work, and strict claim boundaries instead of presenting the workflow as perfectly linear.

How staggered independent lanes worked

How staggered independent lanes worked



I directed the research goals and standards. ChatGPT designed the staged orchestration and evidence rules. Codex executed the local pipeline. GABRIEL rated eligible earlier documentary evidence; later administrative evidence used deterministic local processing and bounded semantic AI review.

I use this page to document how staggered independent lanes worked and to show how the project changed as evidence, infrastructure, and scale required safer designs. The sequence preserves human direction, AI execution, accepted work, and strict claim boundaries instead of presenting the workflow as perfectly linear.

Representative scaling milestones

Representative scaling milestones



I directed the research goals and standards. ChatGPT designed the staged orchestration and evidence rules. Codex executed the local pipeline. GABRIEL rated eligible earlier documentary evidence; later administrative evidence used deterministic local processing and bounded semantic AI review.

I use this page to document representative scaling milestones and to show how the project changed as evidence, infrastructure, and scale required safer designs. The sequence preserves human direction, AI execution, accepted work, and strict claim boundaries instead of presenting the workflow as perfectly linear.

Errors, failures, and repairs

Errors, failures, and repairs

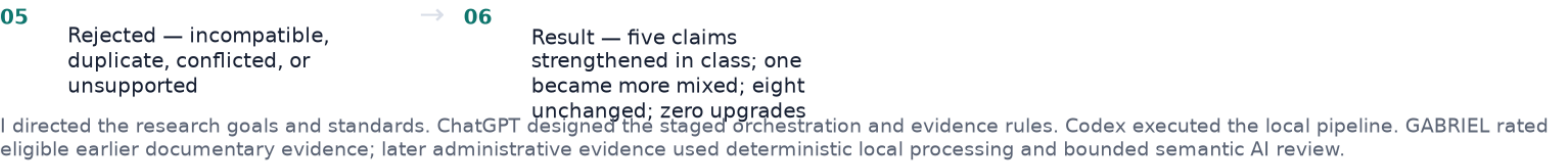


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I use this page to document errors, failures, and repairs and to show how the project changed as evidence, infrastructure, and scale required safer designs. The sequence preserves human direction, AI execution, accepted work, and strict claim boundaries instead of presenting the workflow as perfectly linear.

Strict and broader evidence stayed separate

Strict and broader evidence stayed separate



I use this page to document strict and broader evidence stayed separate and to show how the project changed as evidence, infrastructure, and scale required safer designs. The sequence preserves human direction, AI execution, accepted work, and strict claim boundaries instead of presenting the workflow as perfectly linear.

Status outcomes and technical boundaries

Classification outcomes are kept separate from compensation mechanisms, and underlying tags and counts remain visible.

Unclassified or no direct compensation outcome

None Identified

The retained record documents compensation activity but does not identify a specific wage-setting process. Corrected map-event count: 2,001. This is a classification outcome, not a wage-setting mechanism.

No Direct Compensation Outcome

An administrative event for which the retained record does not show a direct pay outcome. Corrected map-event count: 544. This is a classification outcome, not a wage-setting mechanism.

Unclear

The record points to a compensation event but does not identify the mechanism clearly. Corrected map-event count: 140. This is a classification outcome, not a wage-setting mechanism.

Technical counts and boundaries

Corpus scale

15,163 unique PDFs · 1,029,482 unique native PDF pages · 8,718 substantive HTML documents · 96,484 HTML tables · 1,017,511 HTML rows · 132,188 embedded structured records.

Mechanism layers

2,998 root implementation events in the foundation; 13,391 raw mechanism-link rows; 11,698 corrected map events at the declared unit; 11,692 separate unique administrative mechanism-event IDs in the strict mathematical linkage layer.

Claims and exclusions

14 final claims · 7 counterexamples · 201 unresolved high-impact conflicts · zero compatible external wage matches · zero external growth pairs · no regression · no causal estimate.

Source, method, and handoff notes

Source use

This revision used existing local documentary, administrative, mathematical, review, adjudication, and visual artifacts only. No hosted search, GABRIEL/API call, OCR, redownload, held-source processing, regression, or claim readjudication occurred.

Original atlas

The 76-page first atlas remains archived and byte-preserved. This revision replaces its dashboard prominence only after QA and deployment validation.

Next handoff step

The source library will stream canonical sources directly into bounded compressed split volumes. It will not create a complete uncompressed staging copy and will not assume an external drive.